

ANSWERS & EXPLANATIONS
GENERAL STUDIES (P) TEST – 6333

- Q 1. C** • The expression ‘Public Interest Litigation’ has been borrowed from American jurisprudence, where it was designed to provide legal representation to previously unrepresented groups.
- **Public interest Litigation (PIL)** is litigation filed in a court of law, for the protection of “Public Interest”. Any matter where the interest of the public at large is affected can be redressed by filing a Public Interest Litigation in a court of law. Public interest litigation is not defined in any statute or in any act.

Some of the matters which are entertained under PIL are:

- Bonded Labour matters
- Neglected Children
- Non-payment of minimum wages to workers and exploitation of casual workers
- Atrocities on women
- Environmental pollution and disturbance of ecological balance
- Food adulteration
- Maintenance of heritage and culture

Any citizen can file a public case by filing a petition:

- **Under Art 32 of the Indian Constitution, in the Supreme Court.**
- Under Art 226 of the Indian Constitution, in the High Court.
- **A Public Interest Litigation can be filed against a State/ Central Govt., Municipal Authorities, and not any private party.** The definition of State is the same as given under Article 12 of the Constitution and this includes the Governmental and Parliament of India and the Government and the Legislature of each of the States and all local or other authorities within the territory of India or under the control of the Government of India.
- **Hence, statements 1 and 2 are correct.**

- Q 2. B** • The Doomsday Clock is a symbolic representation rather than a technical or predictive scientific instrument. It is designed to communicate humanity’s proximity to existential global risks, which include not only nuclear threats but also climate change, disruptive technologies, and geopolitical instability. **Hence, option (b) is the correct answer.**
- The Doomsday Clock does not function as a real-time scientific measuring tool based on empirical data. **Hence, option (a) is not correct.**
 - The clock is neither a body nor a mechanism of the United Nations. **Hence, option (c) is not correct.**
 - The International Atomic Energy Agency has no role in the creation, maintenance, or updating of the Doomsday Clock. **Hence, option (d) is not correct.**

- Q 3. C** • Upon launch, a satellite or spacecraft is usually placed in one of several specific orbits around Earth, or it may be sent on an interplanetary journey, meaning it no longer orbits Earth, but instead orbits the Sun.

Geostationary orbit (GEO):

- Satellites in geostationary orbit (GEO) fly above Earth's equator, moving from west to east, exactly matching Earth's rotation: taking 23 hours 56 minutes and 4 seconds to complete one full orbit, i.e. the duration of a sidereal day. This makes GEO satellites appear 'stationary' over one fixed spot. **To keep pace with Earth's spin, they travel at about 3 km per second at an altitude of 35 786 km, much farther than most satellites.**
- **GEO is ideal for satellites that need to stay fixed above a specific location, such as telecommunication satellites, allowing antennas on Earth to stay in a constant position, always pointing at the satellite.**
- **A low Earth orbit (LEO) is, as the name suggests, an orbit that is relatively close to Earth's surface. LEO is considered to be under altitudes of 2000 km.** Unlike satellites in GEO that orbit along Earth's equator, LEO satellites can have their orbital planes tilted at various angles. **Hence, Statement-I is correct but Statement-II is not correct.**
- **LEO's close proximity to Earth is useful for several reasons. It is ideal for satellite imaging, as its closeness allows for higher resolution images.** The International Space Station (ISS) also orbits here as the shorter distance makes it easier for astronauts to reach.

- Q 4. C** • **Indian socialism (as a constitutional objective) emphasizes socio-economic equality and welfare-oriented state action.**

Post-1991 market-oriented reforms shifted the economic model, yet socialistic objectives continue through welfare measures and redistributive aims.

- The term "Socialist" was added to the Preamble in 1976 (42nd Amendment); earlier the constitution already contained socialistic principles through DPSPs.
- No country follows any ideology in a rigid form; states often combine approaches pragmatically while retaining constitutional objectives.

How is India exhibiting the goal of Socialism?

- **Mixed economy model.**
- By providing socio-welfare schemes- It is to reduce socio-economic inequalities. Examples- Antyodaya scheme, Annapurna scheme, Subsidies, Affordable housing scheme, etc.
- Graded taxation or **progressive taxation**- Those earning more can pay high taxes.
- **Land reforms**- Land ceiling Act, Abolition of Zamindari system.
- Labor Laws and Workers Rights.
- **Reservation Policies**, Right to Education, Public Health Initiatives, etc.
- **Hence option (c) is the correct answer.**

- Q 5. A** • Large Language Models (LLMs) are a class of artificial intelligence systems that rely on deep learning, particularly transformer architectures. They are trained on massive text corpora and are designed to perform language-related tasks such as translation, summarisation, and question-answering. **Hence, statement 1 is correct.**

- Reinforcement Learning, on the other hand, is a different AI technique in which an agent interacts with an environment and learns optimal actions through a reward–penalty feedback mechanism. This approach is widely used in areas such as robotics, gaming, and recommendation systems, where sequential decision-making is crucial. **Hence, statement 2 is correct.**
- Computer vision, however, does not require Large Language Models. Vision-related tasks such as image classification, object detection, and pattern recognition are primarily performed using Convolutional Neural Networks (CNNs) and vision transformers. LLMs are therefore not a prerequisite for processing visual data, making any claim that computer vision depends on LLMs scientifically incorrect. **Hence, statement 3 is not correct.**
- Another important AI technique is expert systems, which are rule-based systems that use predefined “if–then” rules and structured knowledge bases. These systems predate modern machine learning and do not learn from large datasets in the way LLMs do. **Hence, statement 4 is correct.**

Q 6. B • The Ring-Fence Policy (1765–1813), primarily implemented by Warren Hastings, was a British East India Company strategy to protect their own territories by establishing a security buffer zone of allied Indian states.

- This policy of Warren Hastings was reflected in his war against the Maratha and Mysore. The chief danger to the Company’s territories was from the Afghan invaders and the Marathas. To safeguard against these dangers, the Company undertook to organise the defence of the frontiers of Awadh on the condition that the Nawab would defray the expenses of the defending army.
- The defence of Awadh constituted the defence of Bengal during that time. Thus, the states brought under the ring-fence system were assured of military assistance against external aggression—but at their own expense.
- In other words, these allies were required to maintain subsidiary forces which were to be organised, equipped, and commanded by the officers of the Company who, in turn, were to be paid by the rulers of these states. Wellesley’s policy of subsidiary alliance was, in fact, an extension of the ring-fence system, which sought to reduce the Indian states into a position of dependence on the British government
- **Hence, option (b) is the correct answer.**

Q 7. D • To tackle the scourge of political defection, in 1969, a committee chaired by Home Minister Y B Chavan examined the issue of defection. It observed that after the 1967 general elections, defections changed the political scene in India as 176 of 376 independent legislators later joined a political party.

- **However, the committee did not recommend any action against independent legislators or any framework for enacting Anti Defection Law.**
- **In the absence of a recommendation on this issue by the Chavan committee, the initial attempts at creating the anti-defection law did not materialize.**
- The next legislative attempt, in 1978, allowed independent and nominated legislators to join a political party once.
- With rising public opinion for an anti-defection law, immediately after securing a clear majority in 1984, the Rajiv Gandhi’s government proposed the new anti-defection bill in the Parliament. After marathon debates, both the Lok Sabha and Rajya Sabha unanimously approved the bill and through

the 52nd amendment to the Constitution of India, the Tenth Schedule was inserted into the constitution. **Hence, statement 1 is not correct.**

The Tenth Schedule contains the following provisions with respect to the disqualification of members of Parliament and the state legislatures on the ground of defection:

Members of Political Parties: A member of a House belonging to any political party becomes disqualified for being a member of the House,

- **if he voluntarily gives up his membership of such political party or**
 - **if he votes or abstains from voting in such House contrary to any direction issued by his political party without obtaining prior permission of such party and such act has not been condoned by the party within 15 days.**
- In the case of 'voluntarily giving up his membership,' the Supreme Court has interpreted that in the absence of a formal resignation by the member, the giving up of membership can be inferred by his conduct. In other judgments, members who have publicly expressed opposition to their party or support for another party were deemed to have resigned.

Thus, a legislator's speech and conduct inside and outside the legislature can lead to disqualification under the anti-defection law. **Hence, statement 2 is not correct.**

- **Independent Members:** An independent member of a House (elected without being set up as a candidate by any political party) becomes disqualified to remain a member of the House if he joins any political party after such election.
- **Nominated Members:** A nominated member of a House becomes disqualified for being a member of the House if he joins any political party after the expiry of six months from the date on which he takes his seat in the House. This means that he may join any political party within six months of taking his seat in the House without inviting this disqualification.

Exceptions

The above disqualification on the ground of defection does not apply in the following two cases:

- If a member goes out of his party as a result of a merger of the party with another party. A merger takes place when two-thirds of the members of the party have agreed to such a merger.
 - If a member, after being elected as the presiding officer of the House, voluntarily gives up the membership of his party or rejoins it after he ceases to hold that office. This exemption has been provided in view of the dignity and impartiality of this office.
- The Anti-Defection Law does not specify a time period for the Presiding Officer to decide on a disqualification plea. Given that courts can intervene only after the Presiding Officer has decided on the matter, the petitioner seeking disqualification has no option but to wait for this decision to be made. **Hence, statement 3 is not correct.**

- Q 8.A**
- India has emerged as the largest emitter of anthropogenic SO₂ in the world, largely because of its heavy reliance on coal for electricity generation. In 2022, India's SO₂ emissions from electricity generation were the highest globally, accounting for over 20 % of total anthropogenic emissions according to recent data. **Hence statement 1 is correct.**
 - Ambient SO₂ levels in many urban monitoring stations have shown a declining trend due to cleaner fuels and pollution control measures. However, overall national emissions remain significant because coal power and heavy industries continue to contribute a large share.

- While Flue Gas Desulfurization (FGD) systems can dramatically reduce SO₂, installation is not universally mandatory without exceptions at present. Recent policy revisions have allowed exemptions or relaxed norms for some coal-fired power plants, delaying full FGD compliance. **Hence statement 2 is not correct.**

Q 9. C • Patents are a record of the technological inventiveness and commercial activities of a nation. They are of essential interest of many engineers and scientists, particularly those in academe who may develop new products and processes and who wish to protect their intellectual property or inventions. Patents in India are governed by the **Patents Act of 1970**, granting 20-year exclusive rights for novel, non-obvious, and industrially applicable inventions.

Only those inventions that fulfil the following three criteria of patentability can be patented:

- The invention must be new/ Novel.
- The invention must be non-obvious/having Inventive Step
- The invention must be useful/having Industrial Applicability

The following are not considered as inventions and are **not patentable under the Patents Act 1970**:

- an invention which is frivolous or which claims anything obviously contrary to well-established natural laws;
- the mere discovery of a scientific principle or the formulation of an abstract theory;
- **the discovery of a living thing or a non-living substance occurring in nature;**
- **a method of agriculture or horticulture;**
- plants and animals in whole or any part thereof other than microorganisms, including seeds, varieties and species and essentially biological processes for production or propagation of plants and animals;
- **a mathematical or business method, a computer programme per se or algorithms;**
- the topography of integrated circuits;
- **inventions relating to atomic energy.**
- **Hence, option (c) is the correct answer.**

Q 10. B • Context: United Nations Environment Programme (UNEP) released Global Status Report for Buildings and Construction (Buildings-GSR) 2024.

Reports released by the United Nations Environment Programme

Emissions Gap Report 2024. Hence, point 2 is correct.

- Tracks the gap between where global emissions are heading with current country commitments and where they ought to be to limit warming to 1.5°C.
- Key Findings:
- GHG emissions rose to a new high of 57 gigatons (Gt) of CO₂
- India ranks 3rd in total GHG emissions [China (1st) & US (2nd)].

Adaptation Gap Report 2024 Hence, point 3 is correct.

- Adaptation Gap is the difference between actually implemented adaptation and a societally set goal, reflecting resource limitations and competing priorities.
- Key Findings:
- Adaptation gap is estimated at US\$187-359 billion per year.

- International public adaptation finance flows to developing countries increased to US\$27.5 billion in 2022.

Global Status Report for Buildings and Construction (Buildings-GSR) 2024 Hence, point 4 is correct.

- Published jointly with Global Alliance for Buildings and Construction (GlobalABC).
- Founded at COP21, it is leading global platform for all built environment stakeholders committed to zero-emission, efficient and resilient buildings and construction sector.
- Key findings
 - Building and Construction Sector (BCS) accounts for -21% of global GHG Emissions.
 - India is attributing 40% of its CO2 emissions to buildings.
- The **Global Landscape of Climate Finance 2024 is primarily a publication produced by the Climate Policy Initiative (CPI)**. It reflects on how climate finance has evolved from 2018 to 2022, exploring both mitigation and adaptation flows across different economies. Despite annual climate finance having more than doubled between 2018 and 2022, a further fivefold increase is required to reach the USD 7.4 trillion needed on average each year through 2030 under the 1.5°C scenario. **Hence, point 1 is not correct.**

- Q 11. D**
- The French East India Company (EIC) was established in 1664. Though last to enter, the French gained considerable ground in India in a short span of time.
 - The first French factory in India was established at Surat in 1668, under the command of François Caron. The French acquired Pondicherry (which was a small fishing village) in 1673 from Sher Khan Lodi, a subordinate of the Sultan of Bijapur, and developed it into their main settlement, but it was not the location of their first factory. **Hence, statement 1 is not correct.**
 - Bombay (Mumbai) was transferred to the British Crown from Portugal as part of the dowry of Catherine of Braganza when she married King Charles II of England in 1661. The Battle of Swally took place in 1612, a significant English victory over the Portuguese which helped the English gain permission to establish a factory in Surat from the Mughals, but it was not directly related to the transfer of Bombay in 1661. **Hence, statement 2 is not correct.**

- Q 12. A**
- **ICAR has developed India's first genome-edited rice varieties – DRR Rice 100 (Kamla) and Pusa DST Rice 1.** These varieties have the potential to bring about revolutionary changes in terms of higher production, climate adaptability, and water conservation. **Hence, statement 1 is correct.**
 - **These new varieties were developed using genome-editing technology based on CRISPR-Cas, which makes precise changes in the organism's genetic material without adding foreign DNA.** **Hence, statement 2 is correct and 3 is not correct.**
 - Gene editing uses site directed nucleases (SDNs) to make changes that may either be a small deletion, a substitution or the addition of a number of nucleotides. Such targeted edits result in a new and desired characteristic. **Gene editing applications are divided into three techniques: SDN-1, SDN-2, and SDN-3. SDN-1 and SDN-2 gene edited plants do not contain introduced new genes, but rather contain small changes in the existing genome of the plant, while SDN-3 plants may contain foreign DNA.** These varieties are edited using edited by using SDN 1 and SDN 2 techniques. **The DRR Rice 100 (Kamala) variety was developed by ICAR-IIRR, Hyderabad, based on Samba Mahsuri (BPT 5204). Its objective is to increase the number of grains**

per panicle and it matures 20 days earlier (~130 days). Due to its shorter duration, it helps save water and fertilizers and reduces methane gas emissions. Its stalk is strong and does not fall.

- **The second variety, Pusa DST Rice 1**, was developed by ICAR-IARI, New Delhi, based on MTU 1010. This variety can increase yields by 9.66% to 30.4% in saline and alkaline soils, with the potential for up to 20% increase in production.

Q 13. B Rainfall and Temperature Patterns:

- **China Type: Found on the eastern margins of continents in subtropical latitudes. It is characterized by a "Modified Monsoon" pattern, summers are hot and humid (maritime air), while winters are cool and dry (continental air).**
- Mediterranean climate (option c) is the exact opposite; it is famous for dry summers and wet winters.
- British type (option a) is cool and temperate with rainfall throughout the year, but lacks the distinct summer maximum or the heat required for silk/mulberry.

Agricultural Indicators:

- **The mention of tea, silk, and mulberry is a major hint. China is the world's leading producer of these, and they thrive in the humid subtropical conditions of South-Central China. Hence, option (b) is the correct answer.**
- Rice is the staple crop here due to the high summer rainfall, distinguishing it from Steppe (option d), which is semi-arid and mostly supports grasses or wheat.

Vegetation:

- Because the climate is a transition between tropical and temperate, the forests are a "mix." Evergreen conifers (like pines) alongside deciduous hardwoods (like oak and camphor) are found.

Q 14. A **Harsha**

- North India splintered into several warrior kingdoms after the downfall of the Gupta Empire. Excepting in the areas that were subdued by the Huns (modern Punjab, Rajasthan, and Malwa), regional identity became pronounced with the emergence of many small states. **Maithriyas** had organized a powerful state in **Sourashtra (Gujarat), with Valabhi as their capital.**
- **Agra and Oudh** were organized into an independent and sovereign state by the **Maukharis**. The **Vakatakas** had recovered their position of ascendancy in the **western Deccan**. Despite political rivalry and conflict among these states, **Thaneswar**, lying north of Delhi between Sutlej and Yamuna, was formed into an independent state **by Pushyabhutis**. It rose to prominence under Harsha. **Harsha ruled** the kingdom as large as that of the Guptas **from 606 to 647 CE.**

Sources: Literary sources

- Bana's Harshacharita
- Hieun Tsang's Si-Yu-ki

Epigraphical sources

- Madhuban copper plate inscription
- Sonpat inscription on the copper seal
- Banskhera copper plate inscription
- Nalanda inscription on clay seals

- Hence option (a) is the correct answer.

Q 15. B As per Section 81 of the Representation of the People Act, 1951, an election petition challenging the result of a Lok Sabha or State Legislative Assembly election must be filed within 45 days from the date of the declaration of the election results.

- This time limit is strictly enforced, and petitions filed after this period are liable to be dismissed by the High Court.
- Hence statement 1 is correct.

According to Section 81 of the Representation of the People Act, 1951, an election petition can be filed by:

- Any candidate who was duly nominated and contested the election.
Any elector who was entitled to vote in the election in the constituency concerned.
 - This means that both candidates and registered voters (electors) in the constituency have the right to file an election petition.
- This provision ensures broader accountability, allowing voters to challenge election results on valid grounds, such as corrupt practices or procedural irregularities.
- Hence statement 2 is not correct.

As per Section 80A of the Representation of the People Act, 1951, the High Court of the state in which the election was held has original jurisdiction to hear and decide election petitions related to Lok Sabha and State Legislative Assembly elections.

- The jurisdiction is typically exercised by a single judge, as assigned by the Chief Justice of the High Court. Appeals against High Court decisions can be made to the Supreme Court under specific conditions.
- Hence statement 3 is correct.

Q 16. D Key provisions of Charter Act of 1833:

- The act elevated the Governor-General of Bengal to the position of Governor-General of India, giving him complete civil and military authority.
- For the first time, the legislation established the government of India with power over the whole British-controlled territory in India.
- It took away the legislative powers of the governors of Bombay and Madras. Exclusive legislative powers were granted to the Governor-General of India for all of British India
- It brought the East India Company's economic activity to a stop, and it became solely an administrative body.
- The Act intended to establish a system of open competition for civil servant selection and stated that the Indians should not be debarred from holding any place, office and employment under the Company. Section 87 of the Charter Act of 1833 provided that no Indian subject of the Company in India was to be debarred from holding any office under the Company "by the reason of his religion, place of birth, descent and colour". Hence option (d) is the correct answer.

Q 17. B Raja Raja I

- Rajaraja assumed a number of titles like Mummidi Chola, Jayankonda and Sivapadasekara. **He was a devout follower of Saivism. He completed the construction of the famous Rajarajeswara temple or Brihadeeswara temple at Tanjore in 1010 A.D. Hence statement 1 is not correct.**

Rajendra I

- His most famous military enterprise was his expedition to north India. The Chola army crossed the Ganges by defeating a number of rulers on its way. Rajendra defeated Mahipala I of Bengal. **To commemorate this successful north-Indian campaign Rajendra founded the city of Gangaikondacholapuram. Hence statement 2 is not correct.**
- He also excavated a large irrigation tank called Cholagangam on the western side of the city.
- Two **famous Uttiramerur inscriptions that give detailed account of the village administration** under the Cholas belong to his reign. **Hence statement 3 is correct.**

Q 18. B Gulf Stream

- The Gulf Stream is a **warm ocean current**.
- It originates in the Gulf of Mexico and flows northward along the eastern coast of North America before crossing the Atlantic.
- It significantly warms the climate of Western Europe.

Benguela Current

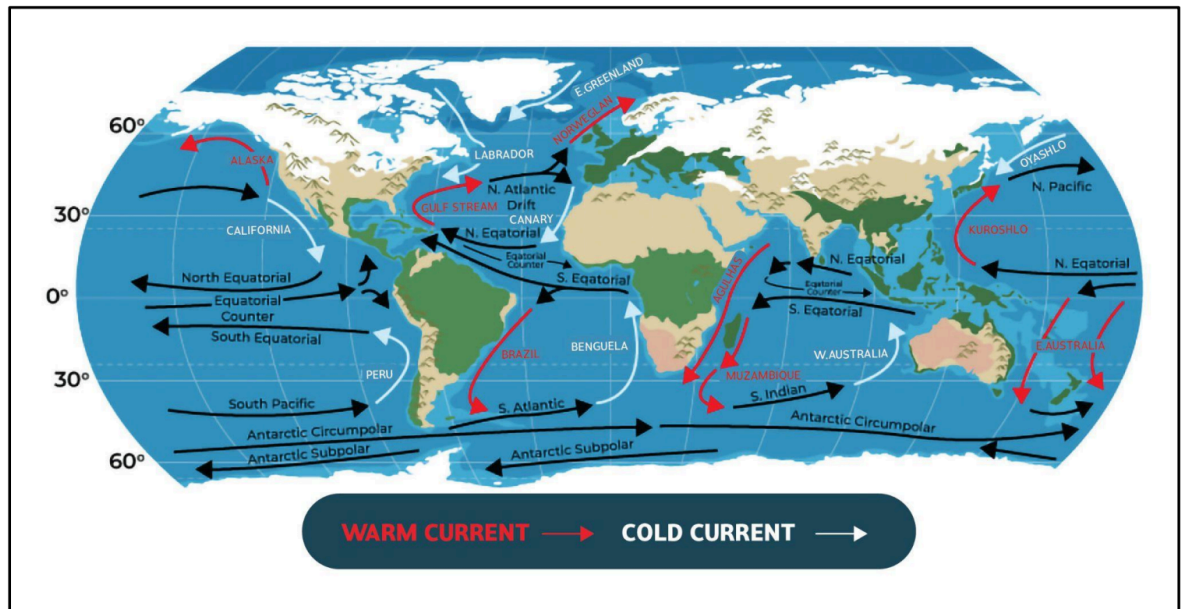
- The Benguela Current is a **cold ocean current**.
- It flows northward along the western coast of southern Africa.
- It is associated with upwelling of cold, nutrient-rich waters, making the region rich in fisheries and contributing to the arid conditions of the Namib Desert.

North Atlantic Drift

- The North Atlantic Drift is a **warm ocean current**.
- It is the northeastern extension of the Gulf Stream.
- This current carries warm waters toward Europe, contributing to mild winters in countries like the UK and Norway.

Labrador Current

- The Labrador Current is a **cold ocean current**.
 - It flows southward from the Arctic region along the eastern coast of Canada.
 - It brings cold water and icebergs and meets the warm Gulf Stream near Newfoundland, causing dense fog.
- **Hence, option (b) is the correct answer.**



- Q 19. C** Kayals are a chain of brackish (slightly salty) lagoons and lakes lying parallel to the Arabian Sea coast. They are formed by the action of rivers and sea waves. **Hence, option (c) is the correct answer.**
- Location: They are the distinguishing feature of the Malabar Coast in the state of Kerala.
 - Vembanad Kayal: This is the largest of these backwaters.
 - Economic & Cultural Importance: These waters are used for fishing and navigation. Every year during the Onam festival, the famous Vallam Kali (Snake Boat Race) is held in these Kayals, specifically in the Punnamada Lake.
- Q 20. B**
- Black carbon is specifically the result of incomplete combustion of carbon-based fuels (biomass and fossil fuels). Fly ash, however, consists largely of the non-combustible mineral matter (inorganic oxides like silicon, aluminum, and iron) present in coal that is carried away by flue gases. **Hence, statement 1 is not correct.**
 - They do not have similar impacts on climate forcing. Black carbon is a strong warming agent (positive forcing). In contrast, fly ash (depending on its composition) often contains "white" or light-colored mineral particles that can reflect sunlight, potentially exerting a cooling effect.
 - Black carbon is officially classified as a Short-Lived Climate Pollutant (SLCP) because it remains in the atmosphere for only a few days to weeks. It is a potent warmer because it strongly absorbs solar radiation and reduces the reflectivity (albedo) of surfaces like snow and ice. **Hence, statement 2 is correct.**

Aspect	Fly Ash	Black Carbon
Basic nature	Inorganic particulate residue	Carbonaceous particulate (soot)
Source process	Complete combustion of coal	Incomplete combustion of fossil fuels & biomass
Major sources	Coal-based thermal power plants	Diesel engines, biomass burning, crop residue burning
Chemical composition	Silica, alumina, iron oxides, calcium	Elemental carbon
Colour	Grey to light brown	Black
Particle size	Mostly coarse to fine PM	Fine to ultrafine PM (PM _{2.5})
Climate impact	Reflects sunlight (cooling effect)	Absorbs sunlight (warming effect)
Role in global warming	Indirect / limited	Strong short-term climate forcer (SLCP)
Health impact	Respiratory irritation, lung diseases	Deep lung penetration, cardiovascular risk
Control measures	Electrostatic precipitators, bag filters	Cleaner fuels, emission standards, cookstove transition
Policy linkage	Fly Ash Utilisation Policy	SLCP reduction, clean air & climate co-benefits

Q 21. A **Hindustan Republican Association (HRA)**

- Genesis: HRA was formed in 1924, as a radical revolutionary organization.
- Objective: To establish a Federated Republic of the United States of India by an organised and armed revolution.
- Founding members: **Ram Prasad Bismil, Ashfaqulla Khan, Sachindra Nath Bakshi, Sachindranath Sanyal, and Jogesh Chandra Chatterjee. Hence, options 1 and 2 are correct.**

Ideologies of HRA:

Socialism: The association envisaged that the "basic principle of the republic shall be universal suffrage and the abolition of all systems which make any kind of exploitation of man by man possible."

- In 1928, Bhagat Singh, Sukhdev, Shiv Verma, Chandra Shekhar Azad and Vijay Kumar Sinha took on the task of reorganizing the HRA and included socialism as a key goal.
- Thus, the HRA was renamed the Hindustan Socialist Republican Association (HSRA).
- Armed overthrow of the imperialist government: The manifesto categorically declared that the foreigners "have no justification to rule over India except the justification of the sword, and therefore the revolutionary party has taken to the sword".
- **Khudiram Bose (December 3, 1889 - August 11, 1908).** He Joined Anushilan Samiti, a revolutionary group in Bengal. Participated in the Muzaffarpur conspiracy (1908): which aimed at targeting British Magistrate Douglas Kingsford. On April 30, 1908, in Muzaffarpur, Bose and Prafulla Chaki bombed the wrong carriage; two British women died. Chaki later committed suicide. Khudiram Bose At age 18, he was arrested, convicted, and hanged on August 11, 1908, much before the formation of HRA. **Hence, option 3 is not correct.**

- Q 22. A • Recent Context:** Iranian officials open to indirect nuclear negotiations with the United States reportedly remain unwilling to make any concessions on Iran's missile program or its role in leading the Axis of Resistance
- The term "Khorasan" originates from the Persian word meaning "Land of the Sun". Historically, Greater Khorasan was a prominent province under various Persian and Islamic empires, covering:
 - Northeastern Iran
 - Parts of Afghanistan (especially Herat and Balkh)
 - Areas of Turkmenistan, Uzbekistan, and Tajikistan
 - In modern times, Khorasan is also mentioned in the context of terrorist organizations such as the ISIS-K (Islamic State – Khorasan Province), indicating their claimed area of operation in the Afghanistan–Pakistan region. **Hence, option (a) is the correct answer.**



- Q 23. A •** A Presidential form of government is a system in which the executive (President) is constitutionally independent of the legislature and is both the head of state and the head of government.

Key Features

- Separation of Powers
 - Direct Election of the President
 - Fixed Tenure
 - Single Executive
 - Non-Membership in Legislature
 - **The President of Indonesia is elected directly by the people using a two-round system. Voters cast ballots for both the President and Vice President as a joint ticket.**
 - **The President is elected by direct universal suffrage in a first-past-the-post system.**
- While South Africa is a republic with a President as both head of state and government, the President is not directly elected by the public.**

- Instead, the National Assembly (lower house of Parliament) elects the President from among its members.
- **Hence option (a) is the correct answer.**

- Q 24. B**
- The Mathura School of Art, flourishing under the Kushan Empire (1st-3rd centuries CE), was a significant indigenous Indian sculptural style using red sandstone, known for its naturalistic yet idealistic figures of Buddhist, Jain, and Brahmanical deities, emphasising spiritual joy and Indian cultural elements, distinct from Gandhara's Greco-Roman style, and laying the groundwork for later Indian art traditions. **Hence, statement 1 is correct.**
 - The Mathura School is known for its assimilative and inclusive character, producing sculptures for all three major Indian religions of the time: Buddhism, Jainism, and Brahmanical (Hindu) traditions. Jaina and Hindu images of the period are carved in the same style, and the images of the Jaina Tirthankaras, or saints, are difficult to distinguish from contemporary images of the Buddha, except by reference to iconography. **Hence, statement 2 is correct.**
 - The Mathura School of Art prominently used local spotted red sandstone for its sculptural works. Blue-grey schist or grey sandstone was the primary material used by the contemporaneous Gandhara School of Art. **Hence, statement 3 is not correct.**

- Q 25. D**
- **National Payments Corporation of India (NPCI)**, an umbrella organisation for operating retail payments and settlement systems in India, is an initiative of **Reserve Bank of India (RBI) and Indian Banks' Association (IBA)** under the provisions of the Payment and Settlement Systems Act, 2007, for creating a robust Payment & Settlement Infrastructure in India. It has been incorporated as a "Not for Profit" Company under the provisions of **Section 25 of Companies Act 1956 (now Section 8 of Companies Act 2013)**. The National Payments Corporation of India (NPCI) operates a comprehensive suite of digital payment and settlement systems to foster a cash-light economy.

Products of NPCI:

- **Unified Payments Interface (UPI)** is a system that empowers users to add multiple bank and other allowed accounts (of any participating bank) into a UPI App, merging several banking features, seamless fund routing & merchant payments under one hood.
- **RuPay** is the first-of-its-kind domestic Card payment network of India, with wide acceptance at ATMs, POS devices and e-commerce websites. It is a highly secure network that protects against anti-phishing.
- **National Payments Corporation of India (NPCI)** has developed the National Electronic Toll Collection (NETC) program to meet the electronic tolling requirements of the Indian market. It offers an interoperable nationwide toll payment solution including clearing house services for settlement and dispute management. Interoperability, as it applies to National Electronic Toll Collection (NETC) system, encompasses a common set of processes, business rules and technical specifications which enable a customer to use their FASTag as payment mode on any of the toll plazas irrespective of who has acquired the toll plaza.
- **Bharat Interface for Money (BHIM)** is an app that lets you make simple, easy and quick payment transactions using Unified Payments Interface (UPI). One can make instant bank-to-bank payments and Pay and collect money using just Mobile number or Virtual Payment Address (UPI ID).

- **IMPS (Immediate Mobile Payment System)** is an instant 24x7 interbank electronic fund transfer service. Customers can use either Mobile Number and MMID OR Account number and IFS Code of the beneficiary to transfer funds using IMPS. It is simple, safe, secure and easy to use, which saves time and cost. IMPS provides robust & real time fund transfer which offers an instant, 24X7, interbank electronic fund transfer service that could be accessed on multiple channels like Mobile, Internet, Branch, ATM and SMS.
- **National Payments Corporation of India (NPCI)** has implemented “National Automated Clearing House (NACH)” for Banks, Financial Institutions, Corporates and Government a web based solution to facilitate interbank, high volume, electronic transactions which are repetitive and periodic in nature. NACH System can be used for making bulk transactions towards distribution of subsidies, dividends, interest, salary, pension etc. and also for bulk transactions towards collection of payments pertaining to telephone, electricity, water, loans, investments in mutual funds, insurance premium etc.
- **National Financial Switch (NFS)** ATM network having 37 members and connecting about 50,000 ATMs was taken over by NPCI from Institute for Development and Research in Banking Technology (IDRBT) in 2009.
- **The Cheque Truncation System (CTS)** is an online, image-based mechanism initiated by the Reserve Bank of India (RBI) that replaces physical cheques with digital images for faster clearing (usually within 24 hours). By utilizing secure electronic transmission, it reduces transit times, enhances security through MICR data analysis, and lowers operational costs.
- **AePS (Aadhaar Enabled Payment System)** is an Indian government initiative by (NPCI) that lets bank customers perform basic financial transactions using their Aadhaar number and biometrics (fingerprint/iris) at Micro ATMs or Business Correspondent (BC) outlets, enabling cash withdrawal, deposit, fund transfers, and balance inquiries, especially in remote areas without needing cards or smartphones.
- **Hence, option (d) is the correct answer.**

- Q 26. C**
- Recent Context: The Minister of External Affairs participated in India’s first-ever engagement in the Weimar Triangle.
 - About Weimar Triangle
 - **It is a regional alliance of France, Germany, and Poland set up in 1991 in the German city of Weimar. He**
 - Initially formed as a key forum for supporting German-Polish reconciliation.
 - Aim: To promote European integration, political dialogue, security cooperation, etc.
 - **NATO: The North Atlantic Treaty Organization includes 32 member states (as of 2024/2025). France (founding member), Germany (joined 1955), and Poland (joined 1999) are all full members of NATO. Hence, option (c) is the correct answer.**
 - **Analysis of other options:**
 - **AUKUS:** An Indo-Pacific security partnership between Australia, the UK, and the USA. None of the Weimar Triangle members are part of this group.
 - **G7:** Consists of Canada, France, Germany, Italy, Japan, the UK, and the USA. While France and Germany are members, Poland is not part of the G7.
 - **ASEAN:** The Association of Southeast Asian Nations is a regional organization for countries in Southeast Asia (e.g., Vietnam, Thailand, Indonesia). None of the Weimar Triangle members belong

to this group as they are European nations

Q 27. A • World Bank finance government programs to support achievement of country development objectives, and support policy and institutional reforms of national and subnational governments by providing budget financing and global expertise.

• **Types of financing by World Bank:**

Types of Financing:

- **Investment Project Financing** provides financing to governments for activities that create the physical/social infrastructure necessary to reduce poverty and create sustainable development. Unlike commercial lending, World Bank IPF not only supplies borrowing countries with needed financing but also serves as a vehicle for sustained, global knowledge transfer and technical assistance.
- **Development Policy Financing (DPF)** provides rapidly disbursing financing to help a borrower address actual or anticipated development financing requirements. DPF supports borrowers in achieving poverty reduction and climate-friendly sustainable and inclusive growth through a program of policy and institutional actions.
- **Program-for-Results** links disbursement of World Bank funds directly to the delivery of defined results, helping countries improve the design and implementation of their own development programs and achieve lasting results by strengthening institutions, enhancing systems, and building capacity.
- **The PRGT is the IMF's main vehicle** for providing concessional financing to low-income countries (LICs). The PRGT's concessional loans support well-designed economic programs that help catalyze additional financing from donors, development institutions, and the private sector. PRGT-supported programs also play a central role in creating the environment for successful debt resolution in distressed countries.
- **Hence, option (a) is the correct answer.**

Q 28. B • Kanheri refers to the Kanheri Caves, a complex of 109 ancient Buddhist caves located in the Sanjay Gandhi National Park in Mumbai, Maharashtra. These rock-cut monuments, carved between the 1st and 10th centuries CE, are significant for their architectural brilliance, intricate sculptures, relief carvings, paintings, and inscriptions, offering a glimpse into India's ancient Buddhist culture and serving as a historical site for study and pilgrimage. **Hence, option (b) is the correct answer.**

Q 29. C • India, known as the 'Land of Spices', is the world's largest producer, consumer, and exporter of spices. Of the 109 spices recognized by the International Organization for Standardization (ISO), India cultivates over 60 varieties. The spice sector contributes approximately 9% to India's total agricultural exports and over 40% of horticultural exports. It retained its position as the leading global exporter, with exports reaching USD 4.45 billion in FY 2024–25. **Hence, statement 1 is correct.**

- **Tobacco Board:** Tobacco is an important commercial crop grown in India. In order to regulate production, promote overseas marketing and control recurring instances of imbalances in supply and demand, the Tobacco Board was established on 1st January 1976 by the Government of India under the Tobacco Board Act of 1975. **The headquarters of Tobacco Board is at Guntur in**

Andhra Pradesh and is headed by a non-executive Chairman appointed by Central Government. **Hence, statement 2 is not correct.**

- **Coffee is cultivated in an area of around 4.90 lakh hectares** predominantly in the traditional areas covering the **States of Karnataka, Kerala and Tamil Nadu**, which contribute to around 97 per cent of the total production and also cultivated to some extent in Non-Traditional Areas of Andhra Pradesh and Odisha and to a lesser extent in the North Eastern States with main emphasis on tribal development and afforestation. **Hence, statement 3 is correct.**

Q 30. D • The **concept of 'equality before law' is of British origin.** As per it the Law is supreme and everyone is equal in the eyes of the Law. In a broader sense, it means that there should be no special privileges in favor of anyone. It also provides for the equal subjection of all persons to the ordinary law of the land administered by ordinary law courts. The state shall not discriminate on the grounds of the status of a person i.e. absence of any special privileges in favour of any person

'The concept of equality before law' is not absolute and there are constitutional and other exceptions to it. These are mentioned below.

- The **President of India and the Governor of States enjoy the following immunities (Article 361)** like they are not answerable to any court for the exercise and performance of the powers and duties of his office; **No criminal proceedings shall be instituted or continued against the President or the Governor in any court during his term of office etc. Hence statement 2 is correct.**
- No member of Parliament (or State Legislature) shall be liable to any proceedings in any court in respect of anything said or any vote given by him in Parliament (or State Legislature) or any committee thereof (Article 105 and Article 194 respectively).
- The **foreign sovereigns (rulers), ambassadors and diplomats enjoy immunity from criminal and civil proceedings. Hence statement 1 is correct.**
- The UNO and its agencies enjoy the diplomatic immunity.
- **No person shall be liable to any civil or criminal proceedings in any court in respect of the publication in a newspaper (or by radio or television) of a substantially true report of an proceedings of either House of Parliament or either House of the Legislature of a State (Article 361-A). Hence statement 3 is correct.**
- Article 31-C is an exception to Article 14. It provides that the laws made by the state for implementing the Directive Principles contained in clause (b) or clause (c) of Article 39 cannot be challenged on the ground that they are violative of Article 14. The Supreme Court held that "where Article 31-C comes in, Article 14 goes out".

Q 31. B • The administrative system of the Delhi Sultanate was centralised and headed by the Sultan. In addition, several departments were administered by powerful nobles or members of the royal family.

- The council of ministers included:
- The **Ariz-i-Mumalik was the head of the military department (Diwan-i-Arz) during the Delhi Sultanate, responsible for recruiting, organising, equipping, and paying the army.** Established by Balban, this official maintained troop readiness and inspected soldiers, with increased importance under Alauddin Khalji.

- This ministry was established to oversee the military organisation. Ariz-i-Mumalik was in command of this department. The ministry was in charge of maintaining the royal contingents, recruiting soldiers, ensuring the army's discipline and fitness, keeping records of every soldier (huliya), examining the horses, and branding them with the royal insignia (dagh). During wartime, Ariz organised military supplies, transportation, and army administration during war, providing constant supplies and being the custodian of war booty. In a hierarchy, after Ariz's position was the Sipahsalar. In addition, Allaudin Khilji introduced the system of dagh (branding), huliya (description), and cash payments to soldiers. This was done to strengthen his authority over the army.
- **Hence, option (b) is the correct answer.**

- Q 32. B**
- The United Nations functions through several principal organs, each playing a distinct role in global governance, and their headquarters locations often reflect their historical and diplomatic significance. **The headquarter of the UN General Assembly is in New York.** It is the central deliberative body where all 193 member states participate equally. It meets at the UN Headquarters complex, established in 1952 as the primary seat of the organization. **Hence pair 1 is correctly matched.**
 - The UN Security Council, also located in New York, is responsible for maintaining international peace and security. Its permanent chamber within the same headquarters symbolizes the concentration of global decision-making power, especially among the five permanent members. **Hence pair 3 is correctly matched.**
 - In contrast, the International Court of Justice (ICJ)—incorrectly paired with Geneva in the question—is actually based in The Hague, Netherlands. The ICJ's location reflects a long-standing tradition of hosting international courts and tribunals in The Hague, often referred to as the world's "legal capital." It serves as the principal judicial organ of the UN, providing advisory opinions and settling legal disputes between member states. **Hence pair 3 is not correctly matched.**

The other organs are:

- UN Secretariat: Administrative organ headed by the UN Secretary-General; carries out day-to-day work of the UN.
- UN Economic and Social Council (ECOSOC) Coordinates economic, social, and environmental work and oversees specialized agencies and commissions.
- UN Trusteeship Council Created to supervise trust territories; suspended operations in 1994 after the last trust territory gained independence (Palau).

- Q 33. B**
- **Context: India has reiterated its commitment to ensuring justice for crimes committed against UN peacekeepers at the meeting of the Group of Friends (GOF).**

Group of Friends

- It is Indian-led initiative and seeks to facilitate the promotion of accountability for all acts of violence against UN peacekeepers.
- Launched in 2022, The Group is a driving force of United Nations Alliance of Civilizations (UNAOC) and plays a vital role in UNAOC's strategic planning and implementation process.

Key Functions:

- Facilitate dialogue among states, UN bodies, and experts.

- Support investigation mechanisms and legal reforms.
- Track implementation of UNSC Resolution 2589 (2021) which mandates accountability for attacks on peacekeepers.
- Advocate for reports from the UN Secretary-General on progress made.
- **Hence option (b) is the correct answer.**

Q 34. C • Pradhan Mantri Anusuchit Jaati Abhyuday Yojana (PM-AJAY) is a centrally sponsored scheme launched by the Ministry of Social Justice and Empowerment targeted at the comprehensive development of Scheduled Castes (SCs). Hence, statement 1 is correct.

- **It aims to promote education, economic empowerment, and infrastructure development for SC communities.**

The **PM-AJAY is an umbrella scheme** that subsumes earlier schemes like: Pradhan Mantri Adarsh Gram Yojana (PMAGY), Special Central Assistance to Scheduled Castes Sub Plan (SCA to SCSP), and Babu Jagjivan Ram Chhatrawas Yojna (BJRCY)

- **Components of the Scheme:**

- **Development of SC dominated villages into an ‘Adarsh Gram’ component.**

- An **‘Adarsh Gram’** is one wherein people have access to various basic services so that the minimum needs of all the sections of the society are fully met and disparities are reduced to a minimum.

- **Developmental Indicators:** Adequate Infrastructure, and Improvement in socio-economic indicators.

- **Funding:** An amount of Rs.21 lakh per village for newly selected villages by the Central Government (scheme is 100% funded by the Central Government; However, the States/UTs are free to provide additional funds).

- **Grants-in-aid for District/State-level Projects for Socio-Economic betterment of SCs**

- **Objectives:** To increase the income of the target population by way of comprehensive livelihood projects

- **Special Provisions for North Eastern States:** 2% of the total scheme budget is reserved for North Eastern States implementing the Scheduled Caste Sub Plan for SCs.

- **Construction/Repair of Educational Hostels for SC students (Boys and Girls)**

- **Objectives:** Containment and reduction of dropout rate of Scheduled Castes Students, especially for SC girls.

- **Eligibility:** Top-ranked Higher Educational Institutions (as per the NIRF) as well as other Central Institutes and State Institutes.

Monitoring and Implementation:

- **Management Information System (MIS):** A centralized portal to capture the data on a real-time basis for each component.

- Independent evaluations through a specialised agency in the area of Rural Development or Social Sciences, or Management, etc.

- **Social Audit:** Done by Gram Sabha at least once a year. **Hence, statement 3 is correct.**

- **The scheme encourages women’s participation but has no strict quota for SC women beneficiaries. Hence, statement 2 is not correct.**

Eligibility under PM-AJAY Scheme:

Adarsh Gram Component:

- **Eligible Villages:** SC population >40% and total population 2500 (latest Census)
- After covering these, villages with ≤40% SC population but a high SC count will also be considered.

- Q 35. B** • The Cripps Mission was a significant diplomatic attempt by the British government in March 1942 to secure Indian cooperation for Britain's efforts in World War II. Led by Sir Stafford Cripps, a member of the British War Cabinet, the mission arrived as Japanese forces were rapidly advancing toward India's borders following the fall of Singapore and Burma.
- The Draft Declaration drafted by Cripps based on his consultations with the Government and different political groups in India stated that **Dominion Status would be granted to India after the war**, and a Constituent Assembly would be set up, with every province having a right to opt out of the body and establish its own union after the drafting. **Hence, statement 1 is not correct.**
 - Further, princely states that did not wish to accede to India could continue in their preexisting relations with the British crown. **Hence, statement 2 is correct.**
 - Mahatma Gandhi met Cripps in Delhi for two hours. He was extremely disappointed with the proposals made by him. Gandhi said to Cripps, "Why did you come if this is what you had to offer? If this is your entire proposal in India, I would advise you to take the next plane home."
 - Gandhi's reaction was sharp, and he was against the draft as it encouraged disunity, infighting and discord, and failed to represent Indian aspirations for true independence. Cripps's mission was a failure as Mahatma Gandhi called the proposal, 'A post-dated cheque on a tottering bank'
- Q 36. C** • **EXIM Bank:** The Export-Import Bank of India (EXIM Bank) is a specialized financial institution established in 1982 to support and promote India's international trade. **It is wholly owned by the Government of India** and operates under the administrative control of the Ministry of Finance. Exim Bank acts as a growth engine for industries and SMEs through a wide range of products and services. This includes import and export of technology, product development, export production, export marketing, pre-shipment export and post-shipment and foreign investment.
- **IIFCL:** IIFCL is a **wholly-owned Government of India company** set up in 2006 to provide long-term financial assistance to viable infrastructure projects through the Scheme for Financing Viable Infrastructure Projects through a Special Purpose Vehicle, broadly referred to as SIFTI. IIFCL has been registered as a NBFC-ND-IFC with the Reserve Bank of India (RBI) since September 2013 and follows the applicable prudential norms of the RBI.
 - **NaBFID:** The National Bank for Financing Infrastructure and Development has been established as an infrastructure focused development financial institution (DFI) under the National Bank for Financing Infrastructure and Development Act, 2021 to support the development of long-term non-recourse infrastructure financing in India including development of bonds and derivatives markets necessary for infrastructure financing and to carry on the business of financing infrastructure. **Currently, it is fully owned by the Government of India.**
 - **Hence, option (c) is the correct answer.**
- Q 37. A** • The *vajrasana* (throne of meditation) at the Mahabodhi temple at Bodh Gaya is a large **stone slab** under the bodhi tree (this pipal tree is supposed to be a descendant of the tree under which the

Buddha attained enlightenment). Its original place and purpose are not known for sure. The 16.5 cm thick *vajrasana* is made of **Chunar sandstone**. Hence statement 1 is correct.

- Its top surface is decorated with a carved geometric pattern that looks like intersecting circles. On the sides are carved floral palmette designs and geese (*hamsas*), in a style **similar to that of the carving on the Ashokan capitals**. Hence statement 2 is not correct.

- Q 38. A**
- The ancient region that encompasses present-day Lucknow and Awadh was a significant part of the Koshala Mahajanapada, one of the sixteen great kingdoms of ancient India. The capital of Koshala was Ayodhya. **Hence, statement 1 is correct.**
 - Lucknow Chikankari embroidery has been granted a Geographical Indication (GI) tag.
 - However, Awadhi Dum Biryani (or Lucknow Biryani) has not been granted a GI tag. Currently, only some specific regional variations of biryani, such as Hyderabad Haleem, have received a GI tag, but biryani itself has not. **Hence, statement 2 is not correct.**
 - The Rumi Darwaza in Lucknow was built by Nawab Asaf-ud-Daulah (Nawab Wazir of Awadh) in 1784, not during Shah Jahan's reign, though it shows Awadhi architecture inspired by Mughal and Turkish styles and uses brick-lime construction common after the Mughals. It was a public works project during a famine, serving as an entrance to Old Lucknow and symbolizing the city, inspired by Istanbul's Sublime Porte.
 - The purpose was to provide employment to the people during the devastating famine of 1784. Mughal Emperor Shah Jahan's reign ended in 1658, over a century before the Rumi Darwaza was built. **Hence, statement 3 is not correct.**

- Q 39. A**
- **Employee Provident Fund (EPF):** The Employee Provident Fund (EPF) is a government-backed retirement savings scheme primarily designed for salaried employees in India. Under this scheme, both the employee and employer contribute a fixed percentage of the employee's salary to the EPF account each month.
 - **National Pension Scheme (NPS):** The National Pension Scheme (NPS) is a voluntary, government-sponsored pension scheme designed to encourage long-term savings for retirement. Unlike EPF, NPS is open to all Indian citizens, including self-employed individuals, between the ages of 18 and 60

• Key Differences Between EPF and NPS

Nature of Returns

- **EPF: Offers guaranteed returns with an interest rate set annually by the government.** The returns are fixed and predictable, making EPF a low-risk investment option.
- **NPS: Provides market-linked returns,** meaning the performance of the NPS is tied to the market. The returns can vary based on the investment choices made within the NPS, such as equity, government bonds, and corporate debt. **Hence, statement 1 is correct.**

Eligibility Criteria

- **EPF: Exclusively available to salaried employees working in the organized sector. Employers with more than 20 employees are mandated to offer EPF to their staff.**
- **NPS: Open to all Indian citizens,** including salaried and self-employed individuals, **aged between 18 and 70.** There is no restriction based on the nature of employment. **Hence, statement 2 is correct.**

Contribution Structure

- **EPF:** Contributions are mandatory for both employees and employers. **The employee contributes 12% of their basic salary, which is matched by the employer.** The entire contribution is automatically deducted from the employee's salary
- **NPS:** The minimum contribution for a National Pension System (NPS) Tier I account is ₹500 per transaction, with a minimum **annual contribution of ₹1,000** to keep the account active. For the optional Tier II account, the minimum initial contribution is ₹1,000, and subsequent contributions are at least ₹250. **Hence, statement 3 is not correct.**

Liquidity and Withdrawal Rules

- **EPF:** Employees can withdraw the entire accumulated corpus upon retirement or after reaching 58 years of age. Partial withdrawals are allowed under specific circumstances such as medical emergencies, marriage, or home purchase.
- **NPS:** Up to 60% of the corpus can be withdrawn as a lump sum at retirement, with the remaining 40% mandatorily invested in annuities. Partial withdrawals before retirement are allowed under certain conditions, like for education, marriage, or buying a house, but these withdrawals are subject to restrictions.

Investment Choices

- **EPF:** EPF investments are managed by the EPFO, with no input required from the employee. The investment is primarily in government securities, ensuring low risk and stable returns.
- **NPS:** Offers a choice of investment options, including equity, government bonds, and corporate debt. Subscribers can choose their preferred investment mix or opt for an auto-choice option where investments are allocated based on the subscriber's age.

Q 40. C **Greenwashing** refers to false, misleading, or exaggerated environmental claims made to create an illusion of sustainability without substantial action. It is commonly used in:

- Corporate advertising
- ESG disclosures
- Product labelling ("eco-friendly", "green", "carbon neutral" without evidence). **Hence option (c) is the correct answer.**
- **Greenhushing:** Deliberately under-reporting or staying silent about genuine environmental efforts to avoid scrutiny, regulation, or accusations of greenwashing.
- **Greenwashing:** Making aspirational or future-oriented sustainability promises without concrete plans or timelines.
- **Greenbleaching:** Removing or diluting sustainability commitments (like targets or disclosures) to escape accountability or legal risk.
- **Greenlabelling:** Using environmental labels or symbols that are self-created, non-standard, or unverified to mislead consumers.

Q 41. B • The earliest remains of Indian architecture are found in Harappa, Mohenjodaro, Ropar, Kalibangan, Lothal, and Rangpur, belonging to the Indus Valley culture or the Harappan culture. Town planning was excellent during this civilization. **Burnt brick was widely used**, roads were wide and at right angles to one another, and city drains were laid out with great skill and forethought. **Hence, statement 1 is correct.**

- Two important remains of the oldest times are fortifications of the old Rajagriha town, in Bihar and the fortified capital of Sisupalgarh, perhaps the ancient Kalinganagar, near Bhubaneswar. The **Rajagriha fortification** wall is made in the roughest possible manner, unhewn stones being piled one on top of the other. This belongs to the 6th-5th century, B.C. However at **Sisupalgarh** in the 2nd-1st century B.C. stone masons were at work using large blocks of stones to make a very well-made fort entrance that could be closed with huge doors turning on hinges. Therefore, the architecture of both places was quite different from each other. **Hence, statement 2 is not correct.**
- **Lomas Rishi** cave in the Barabar Hills of Bihar is an example of early cave architecture. An inscription proves that this was excavated for the Ajivika sect in the time of Ashoka. The entrance of the cave is a representation in **stone** of a hut entrance, with the end of the roof constructed of bent **timber** supported by cross beams, the ends of which are shown protruding. A carved frieze of elephants is a stone imitation of similar work in wood along with a stone imitation of trellis work made of small sticks of bamboo. This is an excellent example showing the development of earlier shapes in timber translated into stone. **Hence, statement 3 is correct.**

Q 42. C • Eutrophication is the process where a body of water becomes excessively enriched with nutrients, particularly nitrogen and phosphorus, from sources like sewage, agricultural runoff, and fertilizers. This leads to an overgrowth of algae and aquatic plants (algal blooms), which die and decompose, consuming large amounts of dissolved oxygen in the water. This oxygen depletion can create "dead zones" where most aquatic life, such as fish, cannot survive.

Causes

- Human activities (cultural eutrophication): This is the primary cause of rapid eutrophication and includes nutrient runoff from sources like:
 - Fertilizers from agriculture
 - Sewage and septic systems
 - Industrial wastewater
- Natural processes: Eutrophication can also occur naturally over long periods as lakes age and accumulate organic matter and sediment from terrestrial runoff.
- **Hence, option (c) is the correct answer.**

Q 43. A The following features characterise the society of the Later Vedic Period (c. 1000–600 BCE):

Society exhibited social stratification based on varna:

- **Varna System:** Society transitioned from a relatively flexible tribal structure to a rigid, hierarchical four-fold Varna system (Brahmins, Kshatriyas, Vaishyas, and Shudras). While initially functional, it became increasingly hereditary. **Hence, statement 1 is not correct.**
- **Gender:** The status of women significantly declined. They were restricted from participating in political assemblies (Sabha and Samiti), lost inheritance rights, and were often viewed as subordinate to men.

Economy was agricultural economy:

- During the Later Vedic Period, the economy shifted from being primarily pastoral (which was the characteristic of the Early Vedic Period) to a settled agricultural economy. **Hence, statement 2 is not correct.**

- While cattle rearing remained important, agriculture became the main occupation, aided by the use of iron tools (Shyama Ayas) for clearing forests and tilling land.

Religious practices like centrality of sacrifices:

- **Religious life was dominated by elaborate rituals and complex sacrifices (Yajnas), such as the Rajasuya, Ashvamedha, and Vajapeya. Hence, statement 3 is correct.**
- The priestly class (Brahmins) gained immense power because these rituals were required to legitimize kingship and ensure prosperity, making sacrifice the central focus of religious practice.

Q 44. A Even during a National Emergency under Article 352, Article 21 (Right to Life and Personal Liberty) cannot be suspended. After the 44th Constitutional Amendment Act, 1978, it was clearly established that Articles 20 and 21 remain enforceable even during a National Emergency. Hence statement 1 is not correct.

- Prior to this amendment, fundamental rights could be suspended wholesale, but post-1978, the Constitution explicitly protects these two rights, even in exceptional times.

President's Rule, imposed under Article 356, occurs when the President assumes control of a state's administration upon the failure of the constitutional machinery in that state.

- During this period, the President takes over the powers of the state government, including those of the Governor, and the state legislature is either suspended or dissolved.

When a state is under President's Rule, the state legislature is not functional, and the President assumes the legislative powers of the state under Article 357.

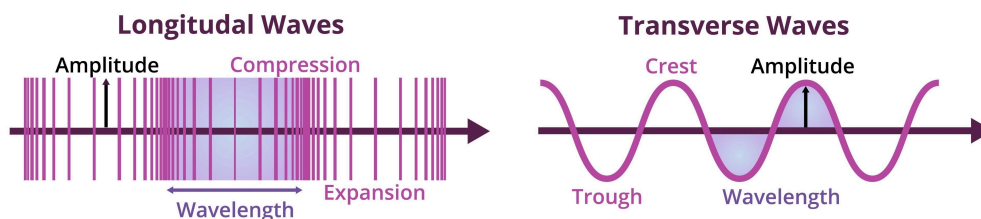
- **If the Lok Sabha is in session, any expenditure from the Consolidated Fund of the State requires parliamentary approval. However, if the Lok Sabha is not in session, Article 357(1) (b) empowers the President to authorize such expenditure temporarily, pending subsequent approval by Parliament when it reconvenes. Hence statement 2 is correct.**
- **Article 360 allows the President, during a Financial Emergency, to order reduction in salaries and allowances of all persons serving the Union or a State, including judges of the Supreme Court and High Courts. There is no exception for the judiciary. Hence statement 3 is not correct.**

Q 45. C • Sound is a mechanical wave created by vibrating objects. These vibrations need to be passed from one particle to another (atom to atom). Without particles to vibrate, such as in a vacuum (outer space), sound cannot travel. This is why "in space, no one can hear you scream." Hence, option (c) is the correct answer.

- **Other options**
- **Option (a):** Sound waves are mechanical waves, not electromagnetic radiation. Electromagnetic waves (like light, X-rays, or radio waves) consist of oscillating electric and magnetic fields and can travel through a vacuum.

Option (b): In air and liquids, sound waves are longitudinal waves, not transverse. In a longitudinal wave, the particles of the medium vibrate parallel to the direction of the wave's travel, creating regions of compression and rarefaction. (Note: Sound can have transverse components in solids, but in the general context of physics, it is classified as longitudinal).

Types of Waves



- **Option (d):** The speed of sound depends on the density and elasticity of the medium. Sound travels fastest in solids, followed by liquids, and slowest in gases (air). This is because molecules in solids are packed tightly together, allowing the vibration to transfer much more quickly.

- Q 46. C**
- **The Private final consumption expenditure (PFCE)** is defined as the expenditure incurred by the resident households and non-profit institutions serving households (NPISH) on final consumption of goods and services, whether made within or outside the economic territory.
 - Since it is convenient to obtain the estimate of total final consumption expenditure made by all households and NPISHs, whether resident or non-resident using commodity flow approach, 'final consumption expenditure in the domestic market' is derived first. This estimate is then adjusted by adding expenditures by residents, incurred abroad (which forms part of imports) and subtracting the expenditures by non-residents, incurred within the economic territory (which forms part of exports), to obtain the estimate 'final consumption expenditure of resident households and NPISHs'.
 - **Any expenditure incurred by residents during their foreign tour/travel will be added in PFCE. However, any expenditure incurred by non-residents and foreign visitors in the domestic market will be deducted from PFCE. Hence, option 1 is added and option 2 is not added.**
 - The consumption expenditure also includes the **imputed gross rent of owner-occupied dwellings**, consumption of own account production evaluated at producers' prices and **payments, in cash and kind, made available to employees. Hence, options 3 is also correct.**
 - **Hence, option (c) is the correct answer.**

- Q 47. A**
- **A derivative is a contract or product that derives its value from an underlying asset.** Derivatives can include a wide range of such assets including indices, currencies, exchange rates, commodities, stocks, or the rate of interest. The buyer and seller of such contracts have opposite estimations of the future trading price. Both the parties bet on the future value of the underlying assets to make a profit.

Depending upon the conditions of a contract, derivatives can be of the following types -

- **Futures** - A futures contract is a legal agreement between two parties to buy or sell the underlying asset at a predetermined future date and price. The contract is executed directly through a regulated and organised exchange.
- **Forwards** - Forward contracts are similar to futures except the deal is not made through an organised or regulated exchange. Since these are Over-The-Counter (OTC) contracts, they carry more counterparty risk for both parties involved.
- **Options** - An options contract gives a trader the right but not an obligation to buy or sell an underlying asset at a predetermined future date and price.

- **Swaps** - A swap is a contractual agreement between two parties to exchange cash flows at a future date based on a pre-planned formula. Similar to forwards, they are OTC contracts and consequently not traded on exchanges.

- **Advantages of derivatives:**

Apart from making trading profitable, other advantages include:

- **Arbitrage advantage:** One can have an arbitrage advantage by buying a commodity at a cheap price in one market and selling it at a higher price in another market.
- **Hedging risk:** With these contracts, one can evade risk. Most times, during trading, risks are likely to rise out from fluctuations in price movement. Investors may want to go with those contracts that are equal to the value of security they own in the market.
- **Enhancing market efficiency and price discovery:** It helps in getting a fair share of the economic value of underlying assets. **Hence, statement 2 is correct.**
- **Derivatives (futures, options, swaps) allow transfer and management of risk without buying or selling the underlying asset (like shares, commodities, or currencies). Hence, statement 1 is correct.**

Disadvantages:

- Derivatives are tools of risk management, **but they often amplify speculation in the market. Hence, statement 3 is not correct.**
- As they are traded in the open market, prices of underlying securities keep on fluctuating. There is exposure to high risk as a result of highly unstable prices of underlying securities.
- There is a counterparty risk when traded OTC fails to undergo a due diligence process.

- Q 48. A**
- **Stockholm Convention:** The Stockholm Convention is a global treaty to protect human health and the environment from POPs (Persistent Organic Pollutants). It was opened for signature in 2001 in Stockholm (Sweden) and became effective in 2004. POPs are listed in various Annexes to the Stockholm Convention after thorough scientific research, deliberations and negotiations among member countries. **Hence, Pair 1 is correctly matched.**
 - **Rotterdam Convention:** The Rotterdam Convention was adopted in 1998 by a Conference of Plenipotentiaries in Rotterdam, the Netherlands. The Convention entered into force on in 2004. It covers pesticides and industrial chemicals that have been banned or severely restricted for health or environmental reasons by Parties and which have been notified by Parties for inclusion in the Prior Informed Consent (PIC) procedure. The Convention creates legally binding obligations for the implementation of the Prior Informed Consent (PIC) procedure. **Hence, Pair 2 is not correctly matched.**
 - **Basel Convention:** It was adopted in 1989 by the Conference of Plenipotentiaries in Basel, Switzerland, the “Basel Convention on the Control of Trans-boundary Movements of Hazardous Wastes and their Disposal”, generally known as the Basel Convention, came into force in 1992. It is an international treaty that aims to reduce the movement of hazardous waste between countries. It particularly focuses on preventing transfer of hazardous waste from developed to less developed countries. **Hence, Pair 3 is not correctly matched.**

- Q 49. A**
- Gene editing tools like CRISPR-Cas9 function by using a guide RNA to direct an enzyme (Cas9) to a specific DNA sequence. This allows for the precise cutting and subsequent modification—such as insertion, deletion, or correction—of genetic material in living cells. **Hence, statement 1 is correct.**

- Introducing entirely new organs in a fully developed organism cannot be achieved by gene editing.

Hence, statement 2 is not correct.

Organ development depends on:

- Embryonic patterning
- Developmental biology
- Cellular differentiation timing

- **Editing DNA alone cannot re-programme adult body architecture.**

Gene-edited microbes are routinely used to produce: **Hence, statement 3 is correct.**

- Insulin
- Vaccines
- Industrial enzymes

Gene editing cannot instantly modify all cells of an adult human body: **Hence, statement 4 is not correct.**

- Delivery of editing tools to every cell is biologically impossible
- Editing occurs cell-by-cell, not system-wide

- Q 50. D • Plantation agriculture is a type of capital-intensive commercial farming practice.** It involves growing cash crops and plants over a large field, for purely commercial purposes. The origin of plantation agriculture can be traced back to around 1500 AD, especially when tropical and sub-tropical regions of the world were colonized by European powers. These regions are ideally suitable for the growth of certain specialized crops, which cannot be grown in the temperate regions and elsewhere, primarily due to climatic factors.
- It is operated on a commercial basis for exporting the entire produce, often with the help of **large-scale enterprises supported with technological inputs**. It also involves post-harvest operations, **including sorting, processing, packaging, and transportation of the produce** for export to international markets worldwide, wherever demand exists. Hence, it is created due to the forces of colonialism, as a part of voyages to explore, colonize, and exploit newer geographical territories carrying definite economic potentials.
- The most important **crops grown in plantation agriculture** can be clubbed under the following categories:
- **Plantation Crops: Cotton, copra, oil palm, rubber, sugarcane, jute, and hemp etc.**
 - **Beverage Crops: Tea, coffee, and cocoa, etc.**
 - **Fruit Crops: Banana and pineapple, etc.**
 - Besides these, certain other crops are also grown in sub-tropical and tropical lands across the continents of the world.
- **Hence, option (d) is the correct answer.**

- Q 51. B • The Bridge Recombinase Mechanism is a gene editing method (DNA editing tool). It exists naturally in our genetic machinery, and can be used to program and edit DNA.** This tool will allow researchers to rearrange, recombine, invert, duplicate, move, and perform other editing operations on very long DNA sequences. In the future, this is expected to lead to more advanced gene editing therapeutics and treatments for diseases. **Hence, statement 1 is not correct.**

- This gene editing method, which exists naturally and has now been discovered, enhances the human ability to edit genomes beyond the capabilities and scope of CRISPR (clustered regularly interspaced short palindromic repeats), a technology that can be used to modify the DNA of living organisms. It utilises mobile genetic elements or “jumping genes”, which cut and paste themselves into genomes and are present in all forms of life, performing on-the-go DNA manipulation through all living beings. **Genetic engineering researchers have discovered that, this tool can be used to edit genes on a larger scale. Hence, statement 2 is correct.**
- Jumping genes are minimal segments of DNA that have the recombinase enzyme, which binds this DNA to other DNA, along with extra DNA segments at the ends of the genes.

Q 52. B • The **Consumer Protection Act, 2019**, is a comprehensive law designed to safeguard consumer interests, ensure fair trade practices, and provide accessible mechanisms for dispute resolution. It applies to both offline and online transactions, including teleshopping, direct selling, and multi-level marketing.

Three-Tier Redressal Mechanism

- The Act establishes Consumer Disputes Redressal Commissions at the district, state, and national levels. These quasi-judicial bodies are tasked with providing simple, speedy, and inexpensive redressal to consumer grievances. **District Commissions hear claims up to ₹1 crore, State Commissions between ₹1 crore and ₹10 crore, and the National Commission above ₹10 crore. Hence, statement 3 is not correct.**
- The Act mandates the establishment of Consumer Disputes Redressal Commissions (CDRCs) at district, state, and national levels. These bodies address complaints regarding unfair trade practices, defective goods or services, and overpricing.
- **District Consumer Disputes Redressal Commission (District Commission):** Established in each district by the state government.
- More than one District Commission can also be established in a district.
- **State Consumer Disputes Redressal Commission (State Commission):** Established in the state by the state government.
- It shall ordinarily function at the state capital and perform its functions at such other places as the state government may notify.
- **National Consumer Disputes Redressal Commission (National Commission):** Established by the central government. It shall ordinarily function at the national capital region and perform its functions at such other places as the central government may notify.
- The central government may also establish regional branches of the National Commission.

Appeal Mechanism

- District to State Commission. **Hence, statement 1 is not correct.**
- State Commission to National Commission.
- Final appeals lie with the Supreme Court. **Hence, statement 2 is correct.**

Q 53. A • The Gandhi-Irwin Pact was signed in March 1931 between Mahatma Gandhi and the then Viceroy of India, Lord Irwin, to bring an end to the ongoing Civil Disobedience Movement. **As part of the agreement, the Congress suspended the Civil Disobedience Movement and agreed to participate in constitutional negotiations. Hence, statement 1 is correct.**

- The British government did not abolish the salt tax entirely. The pact granted Indians the right to produce and collect salt for personal consumption but did not eliminate the tax or the British salt monopoly entirely. Hence, statement 2 is not correct.
- Gandhi agreed to attend the Second Round Table Conference in London, representing the Indian National Congress.

- Q 54. A**
- According to FAO, **Agro-ecological zoning (AEZ)**, defines zones on the basis of combinations of soil, landform and climatic characteristics. Compilation of an AEZ agro-climatic inventory using several climatic variables (e.g., temperature, precipitation, sunshine fraction, relative humidity) gives a general characterization of climatic resources, signifies their suitability for agricultural use, and provides data and indicators related to climatic requirements of crop growth, development, and yield formation.
 - The Penman-Monteith method calculates potential evapotranspiration (PET) by combining an energy balance approach with aerodynamic principles, using meteorological data like net radiation, wind speed, air temperature, and vapor pressure deficit. The Modified Fournier Index (MFI) is a climatological index that measures rainfall erosivity by representing the ratio of average monthly rainfall to average annual rainfall. These are used to estimate the moisture regime, i.e., soil's water content, including the presence or absence of both groundwater and plant-available water, as determined by its physical properties and local climate.
 - Hence, option (a) is the correct answer.

- Q 55. B**
- **Share capital** represents the total amount raised by a company through the issuance of shares to investors. It serves as the backbone of a company's financial structure, enabling it to fund operations, expand business activities, and fulfill long-term objectives. Share capital is divided into various types based on its issuance, subscription, and utilization, each playing a unique role in the company's financial standing.

Various types of Capital:

- **Reserve capital** is that part of the uncalled capital of a company which the limited company has decided by special resolution **not to call except in the event and for the purpose of the company being wound up. Hence, statement 1 is correct.**
- **Authorised, Nominal, or Registered Capital:** This is the **maximum amount of capital a company can raise through public subscription**, as specified in its Memorandum of Association. Companies can increase or decrease this capital based on regulatory procedures. **Hence, statement 2 is correct.**
- **Issued Capital:** Issued capital is a part of authorised capital offered to the public for subscription in the form of shares. Companies may choose not to issue the entire authorised capital at once. Importantly, issued capital cannot exceed authorised capital.
- **Unissued Capital:** This refers to the portion of authorised capital that has not yet been issued. It is essentially the difference between authorised and issued capital.
- **Subscribed Capital:** Subscribed capital is the part of issued capital that the public has accepted and subscribed to. If all shares offered are subscribed, issued capital and subscribed capital become equal.
- **Called-up Capital:** This is the portion of subscribed capital that a company has called upon shareholders to pay. Companies often collect this in installments rather than all at once.

- **Uncalled-up Capital:** This represents the portion of subscribed capital that has not been called for payment yet. It serves as a contingent liability for shareholders.

Paid-up Capital: Paid-up capital is the portion of called-up capital that shareholders have paid. It represents the actual funds received by the company from its shareholders.

- Subscribed capital is the portion of issued capital that investors have agreed to take up, while paid-up capital is the amount actually paid by shareholders. Since subscribers may not have fully paid the amount they subscribed for, **paid-up capital can be less than subscribed capital, but not the other way around. Hence, statement 3 is not correct.**

Q 56. C • Context: India is all set to celebrate its 79th Independence Day on 15th August.

Evolution of the national flag

- 1906: The first national flag designed by Sister Nivedita was hoisted at Parsee Bagan Square in Calcutta, acting as a symbol of resistance against British-made goods during the Swadeshi movement. This flag featured three colours and unique symbols: green top band with eight lotuses, yellow with 'Vande Mataram', and red with a crescent and sun.
 - 1907: The second national flag's version was seen in 1906 when Madam Bhikaji Cama hoisted it at the International Socialist Congress in Stuttgart, Germany. She was seeking support for India's freedom. This flag is also called the Berlin Committee flag. The flag featured eight stars on the top orange band, yellow in the middle with Vande Mataram, and green at the bottom with a crescent moon and star in the corner.
 - 1917: The third flag included diagonal stars. The Home Rule Movement brought the third flag's design in 1917. Annie Besant and Bal Gangadhar Tilak unfurled this new flag, representing India's demand for greater autonomy. This flag included nine horizontal stripes, five red and four green, with the British flag in the corner. It also featured seven diagonal stars, a crescent and star, and a black vertical triangle along the left side.
 - 1921: One of the earliest designs by Pingali Venkayya. This flag was Pingali Venkayya's first design. He showed this to Mahatma Gandhi. The flag included white, green and red stripes with a charkha in the centre, representing India's unity and progress. But the Indian Congress Committee did not adopt this one as the official flag.
 - 1931: This is the second design by Pingali Venkayya. It took a form very similar to the current design, was approved at a Congress Committee meeting, and was officially adopted as the Committee's flag.
 - 1947: The current design's colours remained the same as Pingali's second design, with only a slight modification, replacing the Charkha with the Dharma wheel.
- **Hence option (c) is the correct answer.**

Q 57. B • Operation Sindhu was specifically launched on June 18, 2025, to repatriate Indian citizens (mostly students and workers) caught in the escalating conflict between Iran and Israel. It successfully brought back over 3,500 Indians using a combination of special aircraft and coordinated land routes. **Hence, option (b) is the correct answer.**

Other options:

- **Operation Brahma:** Operation Brahma was a disaster relief and rescue operation undertaken by the Indian government in response to the 2025 Myanmar earthquake.

- **Operation Kaveri:** Launched in 2023 to evacuate Indians from conflict-torn Sudan.
- **Operation Ganga:** Launched in 2022 to rescue Indian nationals (primarily students) from Ukraine following the Russian invasion.

Q 58. A • Brihadeeswarar Temple at Thanjavur temple is a masterpiece of Chola architecture, commissioned by Raja Raja Chola I and completed around 1010 AD. It is a UNESCO World Heritage site and part of the "Great Living Chola Temples". It is renowned for its massive granite vimana (temple tower) and intricate Dravidian style.

- The Madurai Meenakshi Temple is a historic Hindu temple in Madurai, Tamil Nadu, dedicated to Goddess Meenakshi (Parvati) and Lord Sundareswarar (Shiva), renowned for its stunning Dravidian architecture featuring towering, sculptured gopurams (gateways) and intricate carvings, with origins dating back centuries and a complex rebuilt in the 16th-17th centuries, serving as a major pilgrimage site with vibrant festivals like the Chithirai Brahmotsavam. The Meenakshi Temple's foundations were laid by King Kulasekara Pandya.
- The Chennakesava Temple at Belur is a masterpiece of Hoysala architecture and a UNESCO World Heritage site, renowned for its intricate carvings and detailed craftsmanship. Dedicated to Lord Vishnu, the temple was commissioned by King Vishnuvardhana in 1117 CE to celebrate his military victories over the Cholas.
- Virupaksha Temple at Pattadakal: This temple belongs to the Badami Chalukya architecture style, built in the 8th century by Queen Lokamahadevi to commemorate her husband's victory over the Pallavas. The architecture is a prominent example of the Southern (Dravida) style, which later served as a model for the famous Kailasa temple at Ellora.
- **Hence, option (a) is the correct answer.**

Q 59. D • Recent Context: Minamata Convention agrees to end the use of dental amalgam by 2034. The decision was taken at the sixth meeting of the Conference of the Parties to the Minamata Convention on Mercury (COP-6) in Geneva to reduce mercury pollution.

About Mercury:

- Mercury (Hg) is a naturally occurring, heavy, silvery-white transition metal with atomic number 80.
- It is ductile, malleable, and is able to conduct heat and electricity.
- It is the only common metal that is liquid at room temperature.

Sources:

- Natural: Include volcanic eruptions, emissions from the ocean, etc.
- Anthropogenic: Mining (e.g., during gold mining), fossil fuel combustion, metal and cement production, etc.

- **Usage: Historically used in thermometers, barometers, fluorescent lighting, certain batteries, and dental amalgams. Hence, option (d) is the correct answer.**

Q 60. D • Aligarh Scientific Society (1864): Founded by Sir Syed Ahmed Khan in April 1864 at Aligarh to promote modern education, science, and the translation of Western works into Urdu, aiding the intellectual empowerment of Indian Muslims.

- **Indian National Congress (1885):** Founded on December 28, 1885, at Gokuldas Tejpal Sanskrit College in Bombay by A.O. Hume, marking the beginning of organised, nationalist political activity in India.
- **Publication of Hind Swaraj (1909):** Written by Mahatma Gandhi in 1909 while travelling from London to South Africa, this book outlined his concept of self-rule, non-violent resistance, and his critique of modern civilisation.
- **Home Rule League(1916):** Founded in September 1916 in Madras (Chennai), this movement aimed for self-government within the British Empire. It operated alongside Bal Gangadhar Tilak's league (April 1916) to mobilise mass support, playing a key role in the national freedom struggle.
- **Hence, option (d) is the correct answer.**

Q 61. C • The Heron is a Medium Altitude Long Endurance (MALE) UAV developed by Israel Aerospace Industries (IAI). It is a mainstay of the Indian Armed Forces for long-range surveillance and intelligence gathering. **Hence, pair 1 is correctly matched.**

- Rustom II, now officially known as TAPAS-BH-201, is an indigenous MALE UAV designed by DRDO (ADE). It is intended for multi-mission roles, including "round-the-clock" (all-weather) surveillance. **Hence, pair 2 is correctly matched.**
- Indrajaal is India's first indigenous autonomous wide-area counter-drone system. Developed by Grene Robotics, it uses AI to detect, track, and neutralize drone threats over large areas (up to 4,000 sq. km). **Hence, pair 3 is correctly matched.**

Q 62. A • **The Herfindahl-Hirschman Index (HHI)** measures the size of companies relative to the size of the industry they're in. **The HHI is a common measure of market concentration.** It calculates the amount of competitiveness by squaring the market share of each firm competing within a market and then summing the resulting numbers. It can range from close to zero to 10,000 with **lower values indicating a less concentrated market.** It's often used in pre- and post-merger and acquisition (M&A) transactions. **Hence, option (a) is the correct answer.**

- Market concentration has climbed steadily across major sectors in India over the past decade, **but aviation stands out with the highest level of dominance, as measured by the Herfindahl-Hirschman Index (HHI).** The aviation sector's HHI surged to 4,500 in FY25, more than double the threshold for a "highly concentrated" market under standards used by the antitrust division of the US Justice Department and the European Commission.

Q 63. B • Given: Longitude of Chennai = 80° E

- Longitude of Ahmedabad = 72° E
- The longitude difference $80^{\circ} - 72^{\circ} = 8^{\circ}$
- Earth rotates 360° in 24 hours, this means $15^{\circ} = 1$ hour and $1^{\circ} = 4$ minutes
- So, $8^{\circ} \times 4$ minutes = 32 minutes
- **Hence, option (b) is the correct answer.**

Q 64. C • Nayakas were military chiefs who usually controlled forts and had armed supporters. The amara-nayakas were given territories to govern by the raya. They collected taxes and other dues from peasants, craftspersons and traders in the area.

- **Statement 1 is incorrect:** The amara-nayakas sent tribute to the king annually and personally appeared in the royal court with gifts to express their loyalty. Kings occasionally asserted their control over them by transferring them from one place to another. However, during the course of the seventeenth century, many of these nayakas established independent kingdoms. This hastened the collapse of the central imperial structure. Many nayakas submitted to the authority of the kings of Vijayanagara but they often rebelled and had to be subdued by military action.
- **Statement 2 is correct: The amara-nayaka system was a major political innovation of the Vijayanagara Empire.** It is likely that many features of this system were derived from the iqta system of the Delhi Sultanate.
- **Statement 3 is correct:** They retained part of the revenue for personal use and for maintaining a stipulated contingent of horses and elephants. **These contingents provided the Vijayanagara kings with an effective fighting force with which they brought the entire southern peninsula under their control.** Some of the revenue was also used for the maintenance of temples and irrigation works.

- Q 65. C
- Tripura shares its international boundary only with Bangladesh, not Myanmar.
 - Arunachal Pradesh shares a long eastern boundary with Myanmar (as well as China and Bhutan).
 - Nagaland shares a land boundary with Myanmar along its eastern side.
 - Manipur shares an international border with Myanmar (Moreh–Tamu border is well known).
 - **Hence, option (c) is the correct answer.**



- Q 66. A
- Satellite-based communication systems developed through space research play a crucial role in healthcare by enabling telemedicine, remote medical consultations, and emergency health connectivity, especially in remote and rural regions of emerging economies where physical healthcare infrastructure is limited. **Hence, option 1 is correct.**
 - Earth observation and remote sensing technologies further contribute by supporting disease surveillance, including the tracking of vector-borne diseases and epidemics, monitoring environmental determinants of health, and assessing health risks during disasters, thereby strengthening preventive and public health planning. **Hence, option 2 is correct.**
 - Miniaturised sensors and imaging technologies, originally developed for space missions due to strict size, weight, and power constraints, are now widely used in portable diagnostic devices, remote patient monitoring systems, and advanced medical imaging equipment. **Hence, option 3 is not correct.**

- Cryogenic rocket engine technology, while essential for launch vehicles and deep-space missions, has no direct application in healthcare delivery or medical services, making it irrelevant in this context. **Hence, option 4 is not correct.**

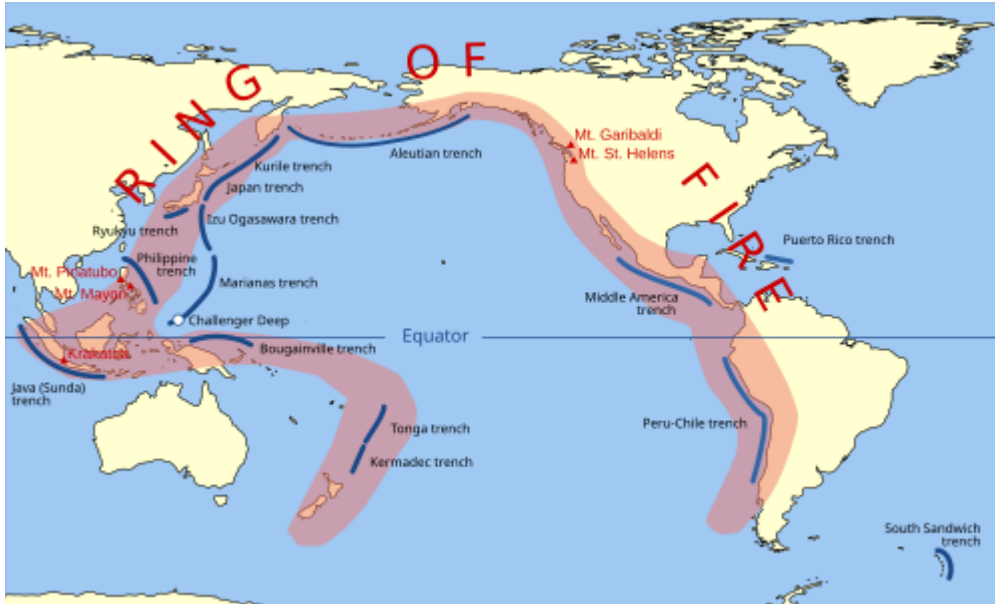
Q 67. D • Context: Recently USA President confirmed that Kazakhstan would be a part of the Abraham Accords.

Abraham Accords

- The Abraham Accords are a **set of agreements that established diplomatic normalization between Israel and several Arab states, beginning with the United Arab Emirates and Bahrain.**
- Named after the biblical Abraham, considered by both Jews and Arabs as their common ancestor and a symbol of brotherhood.
- Announced in August and September 2020 and signed in Washington, D.C. on September 15, 2020, the Accords were mediated by the United States under President Donald Trump.
- The UAE and Bahrain became the first Arab countries to formally recognise Israel since Jordan in 1994.
- In the months that followed, Sudan and Morocco also agreed to normalise relations with Israel, although Sudan's agreement remains unratified as of 2024.
- In July 2025, it was reported that the second Trump administration was seeking to expand the Accords to include Syria, Lebanon, and Saudi Arabia, and in November, it was announced that Kazakhstan agreed to join.
- Its purpose is to defuse tensions in the Middle East by normalising relations between Israel and several so-called moderate Arab states, to ensure formal diplomatic, trade and security ties between these states with Israel.
- **Hence, option (d) is the correct answer.**

Q 68. A • The Pacific Ring of Fire (also known as the Circum-Pacific Belt) is a 40,000 km horseshoe-shaped zone. It is home to nearly 75% of the world's active volcanoes and approximately 90% of the world's earthquakes. It traces the edges of the Pacific Ocean, where the Pacific Plate interacts with many surrounding tectonic plates. Hence, statement 1 is correct.

- The Mid-Atlantic Ridge is located in the Atlantic Ocean, whereas the Ring of Fire is strictly located around the Pacific Ocean. Furthermore, the Mid-Atlantic Ridge is a divergent boundary where plates move apart to create new seafloor, which is a different tectonic process than the subduction typically found in the Ring of Fire. **Hence, statement 2 is not correct.**



- Q 69. A**
- India's commonly found camel is the Dromedary camel (*Camelus dromedarius*). It is a single-humped camel, highly adapted to hot arid conditions, especially the Thar Desert (Rajasthan, Gujarat). This is the dominant camel species used for transport, milk, and livelihoods in western India. **Hence statement 1 is correct.**
 - The Bactrian camel (*Camelus bactrianus*) is a two-humped camel, adapted to cold desert climates. In India, it is found naturally in the Nubra Valley of Ladakh, not in hot deserts. **Hence statement 2 is correct.**
 - Khurai camel is NOT a separate species. It is a breed/strain of the dromedary camel, found mainly in Kachchh (Gujarat).
Its unique ability:
 - Can swim
 - Feeds on mangroves
 - Adapted to marshy and coastal ecosystems. **Hence statement 3 is not correct.**
- Q 70. B**
- All three places Tamralipti (Tamluk in West Bengal), Bharukachchha / Bharuch (Gujarat) and Sopara / Nala Sopara (near Mumbai, Maharashtra), are historically recorded as important maritime and riverine ports and trading centres in ancient India:
 - Tamralipti (Tamluk)** lay near the mouth of the Ganges/Hooghly system and served as a principal eastern gateway for inland Bengal and the Gangetic plain to coastal and overseas trade. It acted as an entrepôt for goods moving between the hinterland and sea routes (including contacts with Southeast Asia and beyond).
 - Bharukachchha (Bharuch / Barygaza)** was a major western port on the Narmada estuary and is frequently referred to in ancient sources as a hub for coastal and overseas trade. It linked Gujarat's produce and manufactures with Arabian Sea maritime routes.
 - Sopara (Nala Sopara)** on the western coast near present-day Mumbai functioned as an active port from early historic times and formed part of the maritime network that connected the Konkan–Western India coast with inland centres and external markets.
 - Hence, option (b) is the correct answer.**

- Q 71. B** • Swami Vivekananda formally established the Ramakrishna Mission on May 1, 1897, to propagate the teachings of his guru, Sri Ramakrishna Paramahansa, and to integrate spiritual life with social service. **Hence, statement 1 is correct.**
- Vivekananda believed that Indian spiritual traditions, particularly Vedanta, were fully rational and compatible with a modern scientific temper. He advocated for a "neo-Vedanta" that balanced Eastern spiritualism with Western material and scientific progress. **Hence, statement 2 is correct.**
 - Vivekananda did not reject social reform; rather, he viewed it as essential for national regeneration. He vehemently opposed the caste system, untouchability, and superstitions, arguing that the upliftment of the masses (both secular and spiritual) was required for India to rise again. **Hence, statement 3 is not correct.**
- Q 72. D** • **The Group of Seven (G7) is an informal bloc of industrialized democracies comprising: France, Germany, Italy, the United Kingdom, Japan, the United States, and Canada. Hence, option (d) is the correct answer.**
- It was formed in 1975 in response to economic and financial cooperation due to energy crisis.
- Q 73. D** • In a first-of-its-kind move, **the International Union for Conservation of Nature (IUCN) has adopted Motion 007 on Soil Security Law, marking the world's first model legal framework aimed at soil protection. Motion 007 has been adopted by The United Nations Convention to Combat Desertification and Food and Agriculture Organization.**
- It defines soil security as “responsible management and preservation of soil to ensure its continued ability to perform vital functions, provide essential ecosystem and biophysical services, and protect against emerging threats to support life on Earth,” and outlines its five dimensions: capacity, condition, capital, connectivity and codification. These dimensions are intended to guide countries, and the world in general, to integrate soil security into policy and programmes and in shaping the potential global convention.
 - **Soil is recognised as the “eighth” existential issue, alongside climate change, food and water scarcity, biodiversity preservation, energy sustainability and environmental maintenance and human well-being.** According to the UN Convention to Combat Desertification (UNCCD). **Hence, Statement-I is not correct and Statement-II is correct.**
- Q 74. C** • **Eighth Five Year Plan (1992-1997):** The eighth plan was postponed by two years because of political uncertainty at the Centre. Worsening Balance of Payment position, rising debt burden widening budget deficits, recession in industry and inflation were the key issues during the launch of the plan. The plan undertook drastic policy measures to combat the bad economic situation and to undertake an annual average growth of 5.6% through introduction of fiscal & economic reforms including liberalisation under the Prime Minister ship of Shri P V Narasimha Rao. **India became a founding member of the WTO in 1995, which falls within the Eighth Plan period. Hence, pair 1 is correctly matched.**
- **Tenth Five Year Plan (2002 - 2007):** Recognising that economic growth cant be the only objective of national plan, **Tenth Plan had set ‘monitorable targets’ for few key indicators (11) of development besides 8 % growth target.** The targets included reduction in gender gaps in literacy and wage rate, reduction in Infant & maternal mortality rates, improvement in literacy, access to potable drinking water cleaning of major polluted rivers, etc. State wise break up of targets for

growth and social development sought to achieve balanced development of all states. **Hence, pair 2 is correctly matched.**

- **Fourth Five Year Plan (1969-1974):** Refusal of supply of essential equipments and raw materials from the allies during Indo Pak war resulted in twin objectives of “ growth with stability “ and “progressive achievement of self reliance “ for the Fourth Plan. The plan introduced the **Tribal Area Development Programme (TADP)**, adopting an area-based and integrated approach for the socio-economic development of tribal regions. During this period, programmes related to agricultural stabilisation, expansion of the Public Distribution System (PDS), promotion of family planning, and strengthening of rural infrastructure and the public sector were also initiated or significantly reinforced. **Hence, pair 3 is correctly matched.**

- Q 75. A**
- Bacteria are prokaryotic cells. They have a defined internal environment filled with cytoplasm, where metabolic activities occur.
 - Viruses are acellular. They consist only of a nucleic acid core (DNA or RNA) inside a protein shell called a capsid. They lack cytoplasm and any internal fluid environment. **Hence, option (a) is the correct answer.**
 - **Other options:**
 - **Option (b):** Viruses are "obligate intracellular parasites," meaning they require a host cell to replicate. Bacteria are generally self-sustaining and replicate independently through binary fission.
 - **Option (c):** Bacteria contain ribosomes (70S type) to synthesize their own proteins. Viruses lack ribosomes entirely; they must hijack the host cell's ribosomes to produce viral proteins.
 - **Option (d):** Neither possesses a "true nucleus." Bacteria are prokaryotes, meaning their genetic material is in a region called the nucleoid without a nuclear membrane. Viruses simply have a genetic strand; they do not have a cellular organization or a nucleus.
- Q 76. B**
- **Co-lending** is an innovative model which involves a **collaborative arrangement where multiple entities partner to extend loans to a borrower** (i.e. individual, group, micro, small and medium enterprise (MSME) or business). Such partnerships have been noted between a traditional bank and NBFCs, housing finance companies (NBFC-HFCs), microfinance institutions (MFIs) or FinTech firms. **Hence, statement 1 is correct.**
 - **The RBI's Co-Lending Model (CLM) allows banks and Registered Non-Banking Financial Companies (NBFCs/HFCs)** to jointly lend to priority sectors, combining lower-cost bank funds with the wider reach of NBFCs. Under this framework, **NBFCs retain at least 20% of the loan risk, while banks fund the remaining 80%**, aiming to enhance credit flow to underserved segments at competitive interest rates. CLM aims to improve the credit flow to the unserved and underserved segments of the economy at an affordable cost. The model works on the principle that banks have lower costs of funds, and NBFCs have greater reach beyond tier-2 centers. **Hence, statement 2 is correct.**
 - **Advantages of co-lending**
 - Co-lending process is beneficial for lenders as well as the borrowers in the following ways:
For borrowers
 - **Borrowers enjoy lower interest rates as banks fund most of the loan compared to the interest rate of a loan solely funded by NBFC. Hence, statement 3 is not correct.**
 - NBFCs provide EMI schemes according to affordability and income.

- Even individuals with moderate credit scores can get access to loans.

For lenders

- Banks can lead to a wider market as NBFCs identify the borrower.
- Co-lending follows an 80:20 funding structure, reducing the bank's exposure to financial risk.
- NBFCs require minimal paperwork so the loan is disbursed quickly.

Q 77. B The Coriolis force is zero at the Equator and maximum at the Poles.”

- The Coriolis force depends on the latitude of a place.
- Mathematically, its magnitude is proportional to the sine of latitude.
- At the Equator (0° latitude); $\sin 0^\circ = 0$; Coriolis force is zero
- At the Poles (90° latitude); $\sin 90^\circ = 1$; Coriolis force is maximum
- This is why large-scale wind systems (like trade winds and westerlies) are clearly deflected away from the Equator. **Hence, statement 1 is correct.**
- The Coriolis force is an apparent force that arises due to the rotation of the Earth on its axis. It has no connection with the Earth's revolution around the Sun. Earth's revolution is responsible for seasons, not for the Coriolis effect. **Hence, statement 2 is not correct.**
- The faster an object moves, the greater the Coriolis deflection it experiences.
This is why:
 - The effect is significant for fast-moving air currents, ocean currents, aircraft, and missiles
 - It is negligible for slow-moving objects like rivers or pedestrians
- **Hence, statement 3 is correct.**

- Q 78. A**
- **Farmana is a significant Harappan archaeological site in Rohtak, Haryana**, famous for its mature Harappan period burial ground with 70 burials dating to approximately 2500–2000 BCE. Excavations, which began in 2006, have revealed an advanced urban layout, a well-planned 26-room complex, and detailed insights into the diet of the period, including the use of spices like ginger and turmeric. **Hence, pair 1 is not correctly matched.**
 - **Kot Diji is a significant Indus Valley Civilization site in Sindh, Pakistan**, known for its early (pre-Harappan, c. 3300 BCE) to mature phases, featuring a fortified citadel and lower town, revealing pottery, kilns, and evidence of extensive burning, crucial for understanding the evolution of the civilization before major sites like Mohenjo-Daro. Excavated by F.A. Khan, it established a key chronological link for the IVC. **Hence, pair 2 is not correctly matched.**
 - **Nageshwar is a crucial Harappan site in Gujarat, India**, famous as a specialized coastal center for producing shell objects like bangles, ladles, and ornaments due to the abundant marine shells available. It functioned as a craft production hub, alongside Balakot, supplying these items to other settlements, indicating extensive Harappan trade and specialized industries away from major cities like Mohenjo-Daro. **Hence, pair 3 is correctly matched.**
 - **Balakot (Kot Bala) is a significant coastal Harappan site in Balochistan, Pakistan**, famous for being a specialized shell craft production center, making bangles and beads from marine shells for trade inland. Excavated by George F. Dales, it shows two main phases: Early Harappan (developmental stages) and Mature Harappan (urban period), revealing its crucial role in the Harappan trade network for shell artifacts, with evidence of large-scale shell working and pre-Harappan settlement. **Hence, pair 4 is not correctly matched.**



- Q 79. D**
- The Indian Himalayan Region (IHR) is much more extensive. It covers 11 States (Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh, Nagaland, Manipur, Mizoram, Tripura, Meghalaya, Assam, and West Bengal) and 2 Union Territories (Jammu & Kashmir and Ladakh). Restricting it to "eight states only" is a common trap in geography questions. **Hence, statement 1 is not correct.**
 - The Valley of Flowers National Park is located in the Kumaon/Garhwal Himalayas in the state of Uttarakhand, not the Kashmir Himalayas. It is a UNESCO World Heritage site known for its meadows of endemic alpine flowers and diverse flora. **Hence, statement 2 is not correct.**
 - Shola forests are not found in the Himalayas. They are the stunted tropical montane forest patches found in the valleys amid rolling grasslands in the higher reaches of the Western Ghats (specifically in Kerala, Tamil Nadu, and Karnataka). The high-altitude grasslands in the Himalayas are actually

called Buggyals (in Uttarakhand) or Marg (in Kashmir, like Gulmarg). **Hence, statement 3 is not correct.**

Q 80. C Clouds

A cloud is a visible mass of minute water droplets or tiny ice crystals formed by the condensation of water vapour at considerable heights in the atmosphere. Because they form at different altitudes, clouds appear in varied shapes, densities, and degrees of transparency. Based on these characteristics, clouds are broadly classified into four main types: cirrus, cumulus, stratus, and nimbus.

Cirrus Clouds

- Found at **high altitudes** (about 8,000–12,000 m).
- Thin, detached, and feathery in appearance.
- **Always white in colour**, giving a delicate look to the sky.
- **Hence, row 1 is correctly matched.**

Cumulus Clouds

- Recognised by their **cotton-wool appearance**.
- **Usually occur at 4,000–7,000 m altitude.**
- Appear in patches scattered across the sky.
- Distinguished by their **flat base**.
- **Hence, row 2 is correctly matched.**

Stratus Clouds

- As the name suggests, these are layered clouds.
- They cover large portions of the sky like a blanket.
- Formed due to loss of heat or the mixing of air masses with different temperatures.

Nimbus Clouds

- Characterised by their **dark grey or black colour**.
- Found at **middle or lower levels**, sometimes so low that they almost touch the ground.
- **Extremely dense and opaque, blocking sunlight.**
- Appear as shapeless masses of thick vapour, often linked with heavy rain.
- **Hence, row 3 is also correctly matched.**

- Q 81. A** • **In mountainous areas, the decrease in temperature with increasing altitude leads to the corresponding change in natural vegetation.** As such, there is a succession of natural vegetation belts in the same order as we see from the tropical to the tundra region. Blue pine and spruce appear at altitudes of 2,225-3,048 m. At many places in this zone, temperate grasslands are also found. But in the higher reaches there is a transition to Alpine forests and pastures. **Silver firs, junipers, pines, birch and rhododendrons, etc. occur between 3,000-4,000 m.** However, these pastures are used extensively for transhumance by tribes like the Gujjars, the Bakarwals, the Bhotiyas and the Gaddis.
- **Western Ghats are locally known by different names such as Sahyadri in Maharashtra, Nilgiri hills in Karnataka and Tamil Nadu and Anaimalai hills and Cardamom hills in Kerala.** Western Ghats are comparatively higher in elevation and more continuous than the Eastern Ghats. **Their average elevation is about 1,500 m with the height increasing from north to south.**

'Anaimudi' (2,695 m), the highest peak of Peninsular plateau is located on the Anaimalai hills of the Western Ghats followed by Dodabetta (2,637 m) on the Nilgiri hills. Hence, Statement I and Statement II are correct, and Statement II is the correct explanation for Statement I.

- Q 82. A** • The Coastal Regulation Zone is the zone near the coastline which includes the coastal area up to 500 metres from the High Tide Line (HTL) and a stage of 100 metres along the banks of streams, estuaries, backwaters, and rivers vulnerable to tidal variations (CRZ).

Classification of the CRZ:

CRZ-I areas are environmentally most critical and are classified as under:

- CRZ-I A: The **ecologically sensitive areas** and the geomorphological features which play a role in the maintaining the integrity of the coast viz. Mangroves; Corals and coral reefs; Sand Dunes; Biologically active Mudflats; Salt Marshes; Turtle nesting grounds; protected areas etc. **Hence pair 2 is not correctly matched.**
- CRZ-I B: The **intertidal zone**. **Hence pair 1 is correctly matched.**
- CRZ-II: The developed land areas up to or close to the shoreline, within the existing municipal limits or in other existing legally designated urban areas.
- CRZ-III: Land areas that are relatively undisturbed (viz rural areas etc) and those do not fall under CRZII. CRZ-III is further classified as:
 - CRZ-III A: Areas with population density more than 2161 per sq km as per 2011 census.
 - CRZ-III B: areas with Student Notes: population density of less than 2161 per sq km, as per 2011 census.
- CRZ- IV: It constitutes the water area and further classified as:
 - CRZ- IV A: The water area and the sea bed area between the LTL up to 12Nm on the seaward side. ◦
 - CRZ- IV B: the water area and the bed area between LTL at the bank of the **tidal influenced water body** to the LTL on the opposite side of the bank, extending from the mouth of the water body at the sea up to the influence of tide, i.e., salinity of five parts per thousand (ppt) during the driest season of the year. **Hence pair 3 is not correctly matched.**
- **Hence option (a) is the correct answer.**

Q 83. B About NAVYA Initiative

- **Ministry:** Joint initiative of the Ministry of Skill Development & Entrepreneurship and the Ministry of Women and Child Development.
- Objective: To provide vocational training to young girls (aged 16-18 years) through PMKVY 4.0.**
 - PM Kaushal Vikas Yojana (PMKVY) 4.0 provides NSQF aligned skill development training including reskilling and upskilling.
- **Aim: To link girls with employment and entrepreneurship opportunities** beyond conventional boundaries, such as Graphic Designer, Smartphone Technician, Drone Assembly Expert, etc.

- Q 84. B** • **At the time of independence around 565 princely states existed in India.** Sardar Patel along with the V P Menon appealed to the patriotic feeling of the rulers along with leveraging Prajamandal

Movement in princely states, to join the Indian dominion in matters of defence, communication and external affairs- these three areas has been part of the paramountcy of the Crown and over which these states had no control.

- **All the princely states were asked to sign two agreements – the instrument of accession and merger agreement.** The instrument of accession would accede a state to India with only matter concerning defence, foreign affairs and communication whereas with the signing of the merger agreement all authority of the state was handed over to the Indian government. The privy purse was an important part of the merger agreement to convince rulers to accede their authority.
- **The privy purse was annual tax-free payment paid to rulers of princely state and their successors from the Consolidated Fund of India.** It was a constitutional right of ruler under article 291 of the Indian Constitution. **It was intended to cover all the expenses of the rulers including those for religious and ceremonial purposes. Hence the statement 2 is correct.**
- **Apart from privy purse they enjoyed various other privileges and recognition under article 362 of the Indian Constitution.** As per this article, the **President of India officially recognised the prince of a state as the ruler. As per the privileges, the Ruler had official titles, could have official vehicles, could host dignitaries.**
- **In 1971, 26th Constitutional Amendment Act was passed to abolish the privileges and privy purse of all the princely states.** This amendment has omitted article 291 and 262 of the Indian Constitution. **The official reason given in 26th amendment act was “the concept of rulership, with privy purses and special privileges unrelated to any current function and social purpose was incompatible with an egalitarian social order....”** Beside **the abolition of the privy purse would reduce the government’s revenue deficit. But this was implicit benefit rather than official reason. Hence statement 1 is not correct.**

Q 85. B Under Article 136, the Constitution of India gives power to the Supreme Court to grant special permission or leave to an aggrieved party to appeal against an order passed in any of the lower courts or tribunals in India. It is referred to as a **special leave petition. Hence option (b) is the correct answer.**

- The Supreme Court is authorized to grant in its discretion special leave to appeal from any judgment in any matter passed by any court or tribunal in the country (except military tribunal and court-martial). This provision contains the four aspects as under:
 - (i) It is a discretionary power and hence, cannot be claimed as a matter of right.
 - (ii) It can be granted in any judgement whether final or interlocutory.
 - (iii) It may be related to any matter—constitutional, civil, criminal, income tax, labor, revenue, advocates, etc.
 - (iv) It can be granted against any court or tribunal and not necessarily against a high court (of course, except a military court).

Curative petition is the last judicial resort available for redressal of grievances in court which is normally decided by judges in-chamber. It is only in rare cases that such petitions are given an open-court hearing.

The concept of curative petition was first evolved by the Supreme Court of India in the matter of Rupa Ashok Hurra vs. Ashok Hurra and Anr. (2002) where the question was whether an aggrieved person is entitled to any relief against the final judgement/order of the Supreme Court, after the dismissal of a review petition.

- The Supreme Court in the said case held that in order to prevent abuse of its process and to cure gross miscarriage of justice, it may reconsider its judgements in the exercise of its inherent powers. For this purpose, the Court has devised what has been termed as a "curative" petition.

Review Petition - A decision of the Supreme Court or of a High Court can be reviewed in a Review Petition. Such a review petition is filed before the same court, generally on very limited grounds, such as an error apparent on the face of the record.

- **Article 137** of the Constitution provides that subject to provisions of any law and rule made under Article 145, the Supreme Court has the power to review any judgement pronounced or order made by it. Under Supreme Court Rules, 2013, such a petition needs to be filed within 30 days from the date of judgement or order. Under these rules, such a review petition can be submitted only on some limited grounds permissible under these rules, since a review petition is not like an appeal.
- The review petition goes to the same bench of judges that delivered the judgement or order sought to be reviewed.
- **Public Interest Litigation** is "a legal action initiated in a court of law for the enforcement of public interest or general interest in which the public or a class of the community have pecuniary interest or some interest by which their legal rights or liabilities are affected

Q 86. D • The event was held in North Goa, India, specifically at Resort Rio, from October 30 to November 27, 2025. **Hence, statement 1 is correct.**

- The FIDE World Cup is a biennial event (held every two years). The previous edition was in 2023, and this edition took place in 2025. **Hence, statement 2 is correct.**
- Javokhir Sindarov won the 2025 title at the age of 19. By doing so, he surpassed previous records to become the youngest player ever to win the FIDE World Cup. **Hence, statement 3 is correct.**

Q 87. B • Recently, **Kozhikode city has been honoured with a membership in the World Health Organization's Global Network of Age-Friendly Cities and Communities (GNAFCC).**

- **GNAFCC** was established in 2010. Its aim is to connect cities, communities and organizations worldwide with the common vision of making their communities great places to grow older in.
- **Hence, option (b) is the correct answer.**

Q 88. A • Tax avoidance involves legally using loopholes or provisions in the tax laws to reduce one's tax liabilities. It is about finding ways to minimize taxes within the framework of the law, even if it might be against the spirit of the law. **Hence statement 1 is correct.**

- Tax evasion involves deliberately misrepresenting one's income or expenses to avoid paying taxes. It is illegal. **Hence statement 2 is not correct.**
- The strategy of a salaried employee living in a rented accommodation despite owning a self-occupied house is used for tax avoidance. An individual can claim House Rent Allowance (HRA) exemption even if they own a house. While technically legal, it can be seen as exploiting a loophole if the primary intention is to avoid tax rather than a genuine need for rented accommodation.
- Depreciation is a legitimate tax deduction that allows businesses to deduct the cost of an asset over its useful life. However, by claiming depreciation beyond the allowed period, the business is artificially inflating its expenses. This reduces its reported profit and consequently its tax liabilities.

The business is misrepresenting the true state of its finances, which is a direct violation of tax law. Hence, this is a practice of tax evasion.

- Business expenses are deductible from taxable income. On the other hand, personal expenses like family travel, personal utilities, etc. cannot be claimed as business expenses. Misclassifying personal expenses as business expenses is a clear case of tax evasion. This is illegal and can result in penalties.

Q 89. A • The two working groups, set up by the Ministry of Agriculture and Farmers' Welfare to review the Pradhan Mantri Fasal Bima Yojana (PMFBY), submitted their recommendations to reverse the falling coverage of the scheme.

- **Pradhan Mantri Fasal Bima Yojana (PMFBY) is a Central Government's flagship crop insurance scheme launched in 2016, replacing the earlier National Agricultural Insurance Scheme (NAIS) and Modified NAIS.**

It aims at supporting sustainable production in the agriculture sector by way of:

- Providing financial support to farmers suffering crop loss/ damage arising out of unforeseen events;
- Stabilizing farmers' income to ensure their continuance in farming;
- Encouraging farmers to adopt innovative and modern agricultural practices; and
- Ensuring credit worthiness of the farmers, crop diversification, and enhancing growth and competitiveness of the agriculture sector besides protecting the farmers from production risks.
- Implementing Agency: Insurance Company selected by the States/UTs from the impanelled list of Insurance Companies through bidding.
- It is a yield index-based scheme and operates on the principle of 'Area Approach' in the selected defined areas called Insurance Units such as a village or Panchayat level.
- **It ensures farmers against all non-preventable natural risks from Pre-sowing to Post-harvest and it is voluntary for all farmers- since Kharif 2020. Earlier, it was compulsory for loanee farmers by the banks. Hence statement 1 is correct and statement 3 is not correct.**
- The Scheme can cover all Food & Oilseeds crops and Annual Commercial/Horticultural Crops for which past yield data is available and for which the requisite number of Crop Cutting Experiments (CCEs) will be conducted being a part of the General Crop Estimation Survey (GCES). **Hence statement 2 is not correct.**

Q 90. B • Alphonse de Candolle (1883) studied 247 cultivated species and his work Origin of Cultivated Plants provided information about the ancestral forms of plants, their origin of domestication and history of cultivated plants.

- The plants of economic value were divided into six classes:

Plants of old world origin: These include

- Plants cultivated for 4000 years and more, e.g. cotton, rice, barley, millet and cotton etc.
- Plants cultivated for 2000 years and more, e.g. alfalfa, oats, radish and yam etc.
- Plants cultivated for less than 2000 years, e.g. **coffee**, okra, artichoke and strawberry etc.

Plants of new world origin

- Plants cultivated for over 2000 years and probably more than 4000 years, e.g. cocoa, maize, sweet potato, tobacco etc.

- Plants cultivated before the time of Columbus, e.g. avocado, **cotton**, pineapple, squash, **potato**, tomato, pumpkin, vanilla, and guava etc.
- **Plants cultivated since the time of Columbus**, e.g. Cinchona, blackberry, **rubber**, blueberry, strawberry, peach etc.
- The potato was first cultivated around 8000–5000 BCE in the Andes Mountains of South America. The cotton plant was cultivated between 5000–3000 BCE in regions such as the Indus Valley and Peru. The coffee plant began to be cultivated around the 9th century CE in the Ethiopia–Yemen region. **The rubber plant was domesticated and brought under cultivation only in the 19th century CE.**
- **Hence, option (b) is the correct answer.**

- Q 91. A**
- The Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Act, 2025, was designed as a "consolidating legislation." It repeals and integrates the provisions of the outdated Atomic Energy Act, 1962 and the Civil Liability for Nuclear Damage (CLND) Act, 2010 into a single, modern framework to facilitate private sector entry and clearer regulatory oversight. **Hence, statement 1 is correct.**
 - While the Act is aimed at a massive expansion of nuclear power, the target of **100 GW is set for the year 2047** (to coincide with Viksit Bharat @2047), not 2030. India's current 2030 targets focus more on reaching approximately 22 GW of nuclear capacity. **Hence, statement 2 is not correct.**
 - The SHANTI Act does open the door for private and foreign participation, but it remains highly regulated. It allows for limited Foreign Direct Investment (up to 49%) in specific areas like equipment manufacturing and construction under the government route, but it does not permit 100% FDI in the operation of nuclear plants, which remains a strategic area under state control (NPCIL) or joint ventures with a majority government stake. **Hence, statement 3 is not correct.**

Key Highlights of the SHANTI Act, 2025:

- **Statutory Regulator:** It grants statutory status to the Atomic Energy Regulatory Board (AERB), making it a more independent body.
- **Graded Liability:** It shifts away from a "one-size-fits-all" liability cap to a Graded Liability Framework where the compensation amount is linked to the risk and thermal capacity of the reactor.
- **Small Modular Reactors (SMRs):** The Act provides a specific legal pathway for the deployment of SMRs by private players to help de-carbonize hard-to-abate industries.

- Q 92. A**
- Context: The five non-permanent members of the UN Security Council (UNSC) elected for two-year terms starting January 1, 2026, are Bahrain, Colombia, the Democratic Republic of the Congo, Latvia, and Liberia. They will serve alongside the five non-permanent members elected in 2024 – Denmark, Greece, Pakistan, Panama, and Somalia – who serve through 2026.
 - The United Nations Security Council, established under the UN Charter in 1945, constitutes one of the UN's six principal organs. It comprises 15 members (5 permanent (P5) and 10 non-permanent members). Out of the 10 non-permanent members, five are elected each year by the General Assembly for a two-year term.
 - The UN General Assembly elects the ten non-permanent members of the Security Council by secret ballot, and a candidate country must receive a two-thirds majority of the votes from members present and voting to win a seat. **Hence, statement 1 is correct.**

- A retiring non-permanent member is not eligible for immediate re-election. This rule is stipulated in the Rules of Procedure of the United Nations General Assembly and is a long-standing provision for non-permanent members of the Security Council. **Hence, statement 3 is not correct.**
- Brazil has served more terms as a non-permanent member of the UN Security Council than India. Brazil has been elected to the Security Council more times than any other UN member state, serving 11 terms by 2022-2023, while India had served 8 terms as of their most recent election in 2020. **Hence, statement 2 is not correct.**

- Q 93. D**
- India's petroleum industry is a comprehensive sector encompassing exploration, production, refining, distribution, and marketing of petroleum and its by-products. This includes upstream activities like extraction of crude oil and natural gas, midstream activities such as transportation and storage, and downstream processes including refining and distribution of fuels like petrol, diesel, LPG, and kerosene. **At present, India has nineteen Public-Sector Undertaking (PSU) refineries, three Private-Sector refineries, and one Joint Venture refinery.** Since there is sufficient refining capacity in the country, **India is a net exporter of petroleum products.** India ranks among the top five refining nations globally, thanks to its robust infrastructure and strategic geographic location. The country is the seventh-largest exporter of refined petroleum products. **Hence, statements 3 and 4 are correct.**
 - **LPG** is sold for both domestic and commercial purposes. In order to prevent misuse of domestic LPG for commercial purpose, the two types of cylinders are coloured differently. For easy identification, cylinders used for domestic use are painted with Signal Red colour and cylinders used for commercial use are painted with Oxford Blue colour. As per The Liquefied Petroleum Gas (Regulation of Supply and Distribution) Order 2000, issued under The Essential Commodities Act, 1955, **a household cannot have more than one LPG connection issued by any of the Public Sector Oil Marketing Company/Companies.** **Hence, statement 1 is correct.**
 - **The prices of petrol and diesel have been made market determined by the Government** and their prices are being decided by the Public Sector Oil Marketing Companies (OMCs) namely, Indian Oil Corporation Limited, Hindustan Petroleum Corporation Limited and Bharat Petroleum Corporation Limited, **in line with the changes in the international market prices and other market conditions.** **Hence, statement 2 is correct.**
- Q 94. B**
- During El Niño, the warm water shifts toward the coast of South America (East Pacific). This causes the air over Australia and Indonesia (West Pacific) to become cooler and drier. Consequently, El Niño is associated with increased incidence of droughts and forest fires in Australia, not a decrease. **Hence, statement 1 is not correct.**
 - The warmer-than-average sea surface temperatures in the Eastern Pacific provide more energy and moisture to the atmosphere. This increases the frequency and intensity of tropical cyclones in that region (near Mexico and the US West Coast) while typically suppressing them in the Atlantic. **Hence, statement 2 is correct.**
 - El Niño generally has a negative impact on the Indian Summer Monsoon. The shift in atmospheric circulation (Walker Circulation) usually leads to "weak" monsoons or drought conditions in India. Conversely, La Niña (the cold phase) is typically associated with a "strengthening" of the Indian monsoon. **Hence, statement 3 is not correct.**

- Q 95. A** • Jet streams are fast, narrow air currents, primarily in the upper troposphere near the tropopause (boundary with the stratosphere), but some weaker jets or related phenomena (like the Somali jet) can dip into the lower stratosphere or lower altitudes, though their core strength and main influence are within the troposphere's upper layers, affecting global weather significantly. **Hence, statement 1 is not correct.**
- The Tropical Easterly Jet (TEJ), which flows from east to west over peninsular India during summer, is a crucial component of the monsoon system. Its presence and strengthening are associated with the formation and intensity of the monsoon trough (Intertropical Convergence Zone) over the Indian region, which enhances monsoon rainfall. **Hence, statement 2 is correct.**
 - Jet streams in the Northern Hemisphere (like the polar front jet and subtropical jet) generally flow from west to east (westerly winds), not east to west, due to the Earth's rotation and the Coriolis effect. The only major exception is the seasonal Tropical Easterly Jet (TEJ) over the Indian Ocean region during the northern summer. **Hence, statement 3 is not correct.**
- Q 96. A** • Most of the world's great deserts (like the Sahara, Atacama, Namib, and the Great Australian Desert) are found between 20° and 30° North and South latitudes on the western sides of continents. This is due to three main factors:
- **Subtropical High-Pressure Belts:** These regions coincide with the Horse Latitudes, where air from the upper atmosphere descends. Descending air is compressed and warms up, which increases its moisture-holding capacity. This prevents condensation and cloud formation, leading to extremely dry, stable atmospheric conditions. **Hence, statement 1 is correct.**
 - **Off-shore Trade Winds:** The Trade Winds blow from East to West (Easterlies). By the time these winds reach the western margins of a continent, they have traveled across vast landmasses and lost all their moisture. These "off-shore" winds blow from the land toward the sea, meaning they bring no rain to the western coast. **Hence, statement 2 is correct.**
- Ocean Currents:** The western coasts of continents in the subtropics are actually washed by Cold Ocean Currents, not warm ones. For example:
- The Canary Current near the Sahara.
 - The Benguela Current near the Namib Desert.
 - The Humboldt (Peru) Current near the Atacama Desert.
 - The West Australian Current near the Great Australian Desert.
 - Cold currents chill the air above them, making it stable and unable to rise and form rain. Warm currents are typically found on the eastern margins of continents. **Hence, statement 3 is not correct.**
- Q 97. D** • **Recent Context:** Recently, a 5-Judge bench of the Supreme Court, in the 16th Presidential Reference (under Article 143), held that the Court cannot impose any timelines for decisions of the President and the Governor on granting assent to Bills under Articles 200/201 of the Constitution. This highlights the advisory nature and limitations of the Supreme Court's role under Article 143.
- Topic in Focus:**
- Article 143 of the Indian Constitution grants the President the power to consult the Supreme Court on matters of law or fact that are of public importance. This provision establishes the advisory jurisdiction of the Supreme Court, allowing the executive to seek an authoritative legal

opinion before making significant decisions, thereby ensuring constitutional propriety and avoiding potential disputes.

Key Aspects:

- **Constitutional Provision:** Article 143 of the Constitution of India deals with the advisory jurisdiction of the Supreme Court.
- **Power to Consult:** The President of India is empowered to seek the opinion of the Supreme Court on any question of law or fact that has arisen or is likely to arise and is of such a nature and of such public importance that it is expedient to obtain such an opinion.
- **Discretion of the Court:** While the President may refer a matter, the Supreme Court is not bound to give an opinion on every reference. It may, in its discretion, refuse to give an opinion.
- **Nature of the Opinion:** The opinion rendered by the Supreme Court under Article 143 is advisory in nature and is not a judicial pronouncement. Therefore, it is not binding on the President or the government.

Purpose: This jurisdiction helps the executive to obtain an authoritative interpretation of legal or constitutional provisions, aiding in policy formulation and governance, and potentially preventing future litigation.

- **Statement 1 is correct:** Article 143 of the Constitution specifically deals with the advisory jurisdiction of the Supreme Court.
- **Statement 2 is correct:** The President is explicitly empowered by Article 143 to seek the Supreme Court's opinion on legal or factual questions.
- **Statement 3 is correct:** The opinion provided by the Supreme Court under Article 143 is advisory and not binding on the President.

- Q 98. B** • **Durand Line: This line is a 2,640-kilometre (1,640 mi) international border between Afghanistan and Pakistan.** It was established in 1893 between British India and the Emirate of Afghanistan. Pakistan inherited the agreement after partition in 1947, and it is internationally recognized as the border, though often disputed by Afghanistan. **Hence, pair 1 is correctly matched.**
- **38th Parallel Line: This line is a popular name given to the latitude 38° N that roughly demarcates North Korea and South Korea.** It was chosen as a temporary division line after World War II, and the Korean Demilitarized Zone (DMZ) now serves as the de facto boundary near this parallel. The border between Indonesia and Malaysia is largely based on geographical features and the remnants of colonial-era borders, particularly on the island of Borneo. **Hence, pair 2 is not correctly matched.**
- **Blue Line: The Blue Line is a demarcation line published by the United Nations on June 7, 2000, for the purposes of determining Israel's withdrawal from Lebanon.** It serves as the de facto boundary separating Lebanon from Israel and the Golan Heights. It closely conforms to the internationally recognized boundary between the two countries, based on the 1949 Armistice Agreement. **Hence, pair 3 is correctly matched.**

- Q 99. C** • The Budapest Convention on Cybercrime (formally known as the Convention on Cybercrime) was opened for signature in 2001 and entered into force in 2004. It remains the first and most significant international treaty seeking to address Internet and computer crime by harmonizing national laws,

improving investigative techniques, and increasing cooperation among nations. **Hence, Statement-I is correct.**

- The Budapest Convention was not drafted under the United Nations (UN). It was drawn up by the Council of Europe (CoE) with the assistance of observer states like the USA, Canada, and Japan. **Hence, Statement-II is not correct.**

- Q 100. B**
- Recently, India added Siliserh Lake in Rajasthan and Kopra reservoir in Chhattisgarh to the List of Wetlands of International Importance (the “Ramsar List”).
 - Siliserh Lake is located in Alwar, Rajasthan, not Haryana. **Hence, pair 1 is not correctly matched.**
 - Kopra Jalashay, which is located in the Bilaspur district of Chhattisgarh. It was recently declared the state's first Ramsar site. **Hence, pair 2 is correctly matched.**
 - Gogabeel Lake is located in the Katihar district of Bihar, not Assam. It is an oxbow lake between the Ganga and Mahananda rivers and is Bihar's first community reserve. **Hence, pair 3 is not correctly matched.**
 - Gokul Jalashay, which is located in the Buxar district of Bihar. It is an oxbow lake on the southern edge of the Ganga River and was also recently designated as a Ramsar site. **Hence, pair 4 is correctly matched.**

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